

Application of Hole-Selective Self-Assembled Monolayers in Inverted Perovskite Solar Cells

Guiqiang Wang,* Kaifei Chen, Long Cheng, Dongsheng Wang, Fanning Meng, and Wanchun Xiang

Hole-transport layer (HTL) is of paramount importance to construct high-performance inverted perovskite solar cells (PSCs) because it not only determines the hole extraction and transport but also influences the quality of perovskite layer. Recently, self-assembled monolayers are adopted as very effective hole-selective layer to construct high-performance inverted PSCs. Compared with conventional HTL, hole-selective self-assembled monolayers (HSSAMs) offer the benefits of minimal material consumption and parasitic absorption, simple and scalable processing, and the versatility in the interface modification. Through molecule design and coating process optimization, the high-quality HSSAMs are obtained, which enable the HSSAM-based inverted PSCs to achieve greatly promoted photovoltaic performance. Herein, the progress of HSSAMs used in inverted PSCs is summarized. First, the structure characteristics of HSSAM molecules are described. Then, the effect of the structure of HSSAM molecules on their function in boosting the device performance and stability is discussed. Furthermore, the deposition strategies to form high-quality HSSAMs for inverted PSCs are analyzed. Finally, the advantages and challenges associated with application of HSSAMs in inverted PSCs are discussed, and the perspectives of the future research trends on HSSAMs for further promoting the performance of inverted PSCs are suggested.

Among photovoltaic technologies, perovskite solar cells (PSCs) stand out and have been getting more and more attention from society because of their low cost, easy fabrication, and skyrocketing efficiency.^[1–3] Through composition engineering, interface engineering, and structure optimization, PSCs have delivered high power conversion efficiency (PCE) exceeding 26%, being close to the efficiency of single-crystal silicon cell.^[4–9]

To date, almost all high-performance PSCs possess the layer-by-layer structure with perovskite layer sandwiched between electron-transport layer (ETL) and hole-transport layer (HTL). Thus, PSCs can be divided into n-i-p (regular) and p-i-n (inverted) junction architectures based on the position of ETL in front end or back end for perovskite layer. Until now, the maximum PCE of inverted PSCs is lower than that of regular devices mainly due to high charge recombination loss and lower open-circuit voltage resulted from the unperfect energy level alignment at the interface of inverted PSCs.^[10–13]

1. Introduction

Photovoltaic cells can directly convert sunlight into electricity and therefore have been considered as one of the most promising technologies to address the issues of the energy crisis and the environmental pollution induced by fossil fuel utilization.

Nevertheless, the inverted PSCs display several unique advantages in comparison with the regular devices, such as much less hysteresis, low-temperature fabricating process, high operational stability, and superior compatibility with tandem solar cells.^[14–17] In addition, the performance gap between the inverted and regular PSCs is gradually reducing through interface engineering, additive engineering, and development of new hole-selective contact.^[18–21] These qualities make the inverted PSCs a promising contender for further development and eventual industrialization and practical deployment.

For inverted PSCs, the perovskite layer is deposited on the HTL. Therefore, the HTL is of paramount importance to fabricate high-performance inverted PSCs because it not only determines the hole extraction from perovskite layer but also influences the quality of perovskite layer.^[22,23] Currently, the most commonly used HTL materials in inverted PSCs in terms of the hole-transport ability and the optical transparency are inorganic metal oxides (such as NiO_x and CuO_x) and organic semiconductors (such as poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS) and poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA)).^[21,24–26] These HTLs should fully cover the transparent conductive oxide (TCO) substrates

G. Wang, K. Chen, L. Cheng, D. Wang, F. Meng
School of Chemistry and Materials
Bohai University
Jinzhou 121003, China
E-mail: wgqiang@bhu.edu.cn

W. Xiang
Key Laboratory for Applied Surface and Colloid Chemistry
Ministry of Education
Shaanxi Key Laboratory for Advanced Energy Devices
Shaanxi Engineering Lab for Advanced Energy Technology
School of Materials Science & Engineering
Shaanxi Normal University
Xi'an 710119, P. R. China

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/solr.202300996>.

DOI: 10.1002/solr.202300996

to avoid the direct contact between perovskite film with the TCO substrate that leads to serious interface recombination, while at the same time, they should be as thin as possible to minimize the parasitic absorption and the resistance loss. This is a dilemma for conventional HTL used in inverted PSCs, which would become bigger as the cell area increases. Furthermore, these conventional HTL materials are not cost-effective because of their time-consuming complicated synthesis process.

Recently, employing the self-assembled monolayer (SAM) as the hole-selective contact in inverted PSCs has been demonstrated to be a facile and effective strategy to circumvent the abovementioned problems derived from the utilization of conventional HTL.^[27,28] Hole-selective self-assembled monolayers (HSSAMs) provide multiple advantages over the conventional HTL, including conformal and uniform coating, minimized thickness, negligible optical and resistance loss, and the versatility in the interface modification. In 2018, HSSAM was first introduced into inverted PSCs, achieving a PCE of 17.8%.^[29] Thereafter, employing HSSAMs to construct inverted PSCs has attracting ever-increasing interest. The molecule-structure design and the coating-process optimization of HSSAMs have resulted in a remarkable improvement in the device performance. The certified PCE of exceeding 25% has been reported for HSSAM-based inverted PSC,^[30] demonstrating that HSSAMs are particularly promising for promoting the photovoltaic performance of inverted PSCs. In this review, we summarize the progress of HSSAMs used in inverted PSCs and highlight the functions of HSSAMs in enhancing the performance of inverted PSCs. The advantages and challenges associated with application of HSSAMs in inverted PSCs are discussed, and the perspectives of the future research trends on HSSAMs for further promoting the performance of inverted PSCs are suggested.

2. The Structure Characteristics of HSSAM Molecules

HSSAMs can be considered as the 2D nanomaterials formed by a thermodynamically favorable self-assembly of molecules on the surface of the substrate.^[31–33] To produce the high-performance inverted PSCs, HSSAMs should function not only as the hole-selective contact to enhance the hole extraction and transport but also as the interface modification layer to improve the perovskite quality and reduce the interface recombination. Therefore, the molecules of HSSAMs should be judiciously designed to meet the above requirements. **Figure 1** summarizes the molecular structure of representative HSSAMs used in inverted PSCs. Typically, HSSAM molecules are designed to contain three parts: 1) the anchoring group that is responsible for connecting HSSAM molecules to the surface of TCO substrates through the chemical attachment. The interaction between the anchoring group and the TCO substrates also determines the coverage of HSSAMs on the substrate surface. Phosphoric acid, carboxylic acid, and boronic acid, which are suitable to bond the TCO substrate, are usually used as the anchoring group of HSSAMs molecules. 2) The head group that is responsible for modulating the surface properties of HSSAMs. Most of HSSAMs used in inverted PSCs adopted the electron-rich heterocycles as the head groups, such as carbazole-based, phenothiazine-based, and

triphenylamine-based derivatives. Their functional groups are designed to interact with perovskite layer and modulate the surface wetting and work function of HSSAMs, aiming to improve the perovskite quality, enhance the hole extraction, and reduce the interface recombination. 3) The spacer group that links the anchoring group and the head group. The spacer group is the backbone of HSSAM molecules and responsible for molecule ordering during the self-assembled process through the lateral noncovalent interaction. The conjugated aromatic ring and non-conjugated alkyl chain are the most commonly used spacer group units. The length of spacer groups defines the thickness of HSSAMs and influences the charge transport.

3. HSSAMs Used in Inverted PSCs

In 2018, HSSAM was employed to fabricate inverted PSCs for first time by Getautis's group.^[29] They synthesized a novel molecule of (2-{3,6-bis[bis(4-methoxyphenyl)amino]-9H-carbazol-9-yl} ethyl)phosphonic acid (V1306) that contains both the phosphonic acid anchoring group and the carbazole-based hole-transporting fragment. V1306 molecules could self-assemble into a SAM on the surface of indium tin oxide (ITO) substrate by an immersing process. The optimized V1306 SAM showed a suitable ionization potential for efficient hole extraction from perovskite film and had a negligible parasitic absorption, which enabled V1306 SAM to be an excellent HSSAM for inverted PSCs. Consequently, the fabricated inverted PSC with a structure of ITO/V1306 HSSAM/perovskite/C₆₀/BCP/Cu delivered a PCE of 17.8%. In the same year, a MC-43 (4'-[bis(2',4'-dimethoxybiphenyl-4-yl)amino]-biphenyl-4-carboxylic acid) SAM was also reported as the hole-selective contact for inverted PSCs, delivering a PCE of 17.3%.^[34] Notably, the short-circuit current density (J_{sc}) of inverted PSCs with HSSAMs was higher than those of the cells with the commonly used PTAA HTL because of the negligible parasitic absorption of HSSAMs. Therefore, these works suggested a novel approach to fabricate high-performance inverted PSCs. Through the molecule-structure design and the coating-process optimization of HSSAMs, the performance of inverted PSCs with HSSAMs has been remarkably improved.

3.1. The Molecular Structure Design of HSSAMs

As discussed previously, HSSAM molecules consist of three fragments: the anchoring group, the spacer group, and the head group. Such molecular structure provides a possibility for reasonable design and synthesis to realize the specific performance required in a certain application.

3.1.1. The Head Groups of HSSAM Molecules

The head group of HSSAMs contacts directly with perovskite layer and therefore plays a key role in controlling the interface properties that are the critical factors affecting the perovskite quality and the interface charge transport and recombination. Thus, the fine design of the head group of HSSAM molecules can provide more opportunity to promote the efficiency and stability of inverted PSCs.

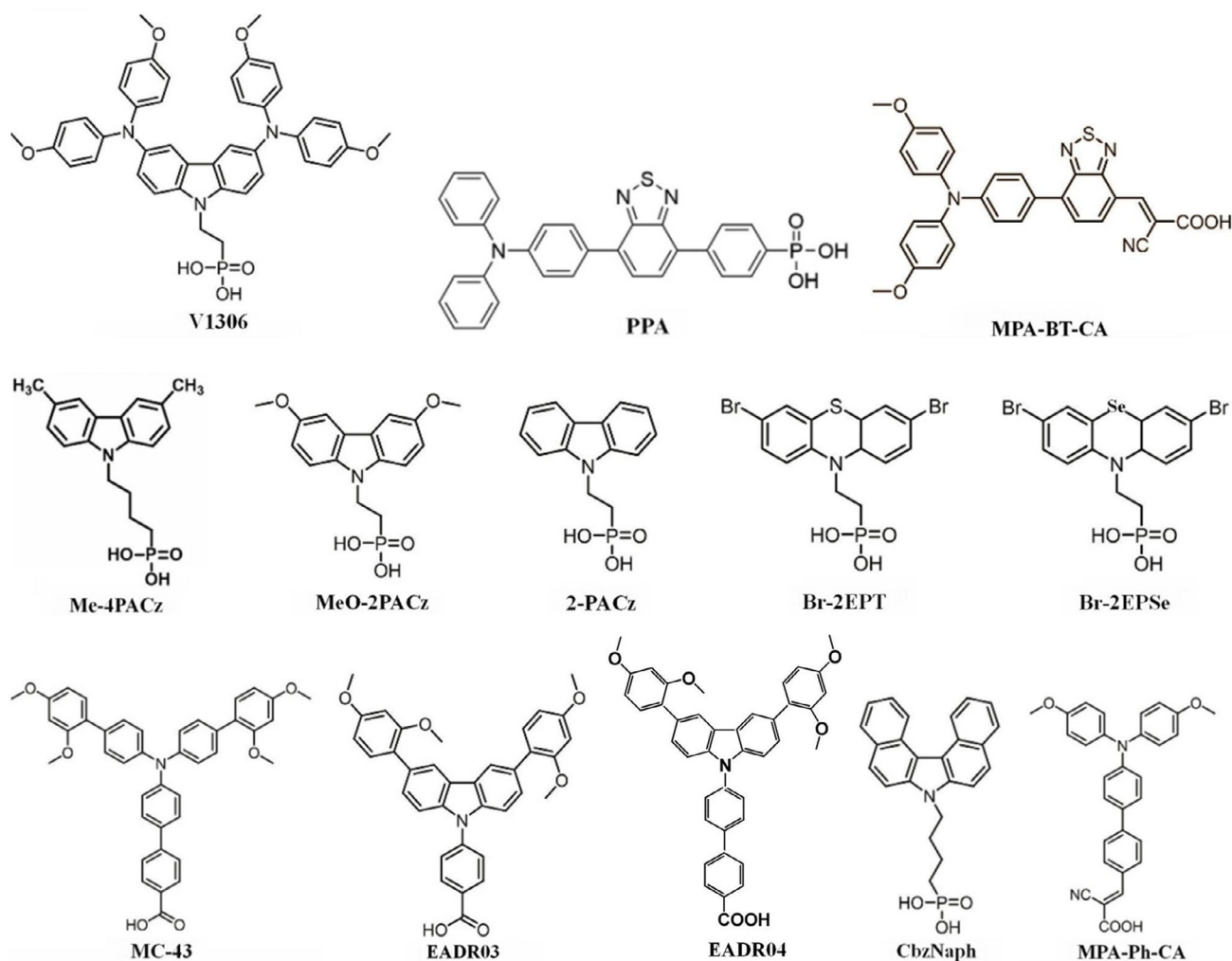


Figure 1. The molecular structure of representative HSSAMs used in inverted PSCs.

The carbazole-based derivatives are the most widely used head groups for HSSAM molecules due to their electron-rich characteristic and large potential for interface passivation.^[33,35] Introducing the substituting group into carbazole moiety can tune the carbazole-based molecule dipole moment and ameliorate the surface wettability of carbazole-based HSSAMs, which contributed to the deposition of compact perovskite film.^[36] Aktas et al. synthesized two carbazole-based molecules with 1,3-dimethoxybenzene substituent for the carbazole moiety, EADR03 (4-(3,6-bis(2,4-dimethoxyphenyl)-9H-carbazol-9-yl)benzoic acid), and EADR04 (4'-(3,6-bis(2,4-dimethoxyphenyl)-9H-carbazol-9-yl)-[1,10-biphenyl]-4-carboxylic acid). Both EADR03 and EADR04 could self-assemble on the surface of ITO through the condensation reaction between the carboxylic acid anchoring group and the hydroxyl group at the ITO surface to form HSSAMs. EADR03 and EADR04 HSSAMs could passivate the interface and extract holes efficiently from the perovskite film, which enabled the assembled inverted PSC with wide bandgap perovskite (1.63 eV) to deliver a PCE of 21%.^[37]

The methoxyl substituent is prevalent in the hole-transport materials of regular PSCs due to their passivation function at

the interface with perovskite layer.^[38,39] Thus, the methoxyl-substituted carbazole-based molecules are also a good choice to form HSSAMs for inverted PSCs. Bakr et al. employed a [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) molecule to form the HSSAM on ITO substrate.^[40] MeO-2PACz HSSAM enhanced the wettability of the substrate, resulting in the growth of large-area $\text{FA}_{0.6}\text{MA}_{0.4}\text{PbI}_3$ perovskite single crystal atop it. The methoxyl group of MeO-2PACz could strongly bond with perovskite layer, passivating the interface defects^[41,42] and suppressing the delamination of perovskite layer from the substrate. The assembled inverted PSC with an ITO/MeO-2PACz HSSAM/ $\text{FA}_{0.6}\text{MA}_{0.4}\text{PbI}_3$ single crystal/ C_{60} /bathocuproine (BCP)/Cu structure achieved a high PCE of 23.1%.

Nevertheless, the introduction of the methoxyl substituent to carbazole moiety greatly decrease the dipole moment of HSSAM because of the compensation of the polarization charge in the anchoring group by the methoxyl groups on top of carbazole moiety.^[43–46] Therefore, the shift in the work function (WF) between the pristine and the HSSAM-covered ITO substrates is higher for ([2-(9H-carbazol-9-yl)ethyl]phosphonic acid)

(2PACz) as compared to MeO-2PACz.^[44–46] Clearly, as shown in **Figure 2a**, 2PACz HSSAM is energetically more hole selective than MeO-2PACz HSSAM, resulting in a more efficient hole extraction and faster hole transport. The introduction of methyl substituent hardly changed the molecule dipole moment. Albrecht et al. reported that the application of a methyl group substituted carbazole-based molecule, Me-4PACz ([4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid), to form HSSAM in inverted PSCs led to a more matched energy level alignment with the valence band edge of perovskite compared

to the application of 2PACz HSSAM (**Figure 2b**). In addition, Me-4PACz molecules could efficiently passivate the interface and reduce the interface defects.^[46] As a result, the inverted PSCs fabricated with Me-4PACz HSSAM and wide-bandgap (1.68 eV) perovskite film delivered a high PCE of 20.8% with a fill factor (FF) of 82% (**Figure 2c**). It should be noted that the efficiency and stability of inverted PSCs with MeO-2PACz, 2PACz, and Me-4PACz HSSAMs are all better than those of the device with conventional PTAA HTL, demonstrating the superiority of HSSAMs over the conventional HTL in inverted PSCs.

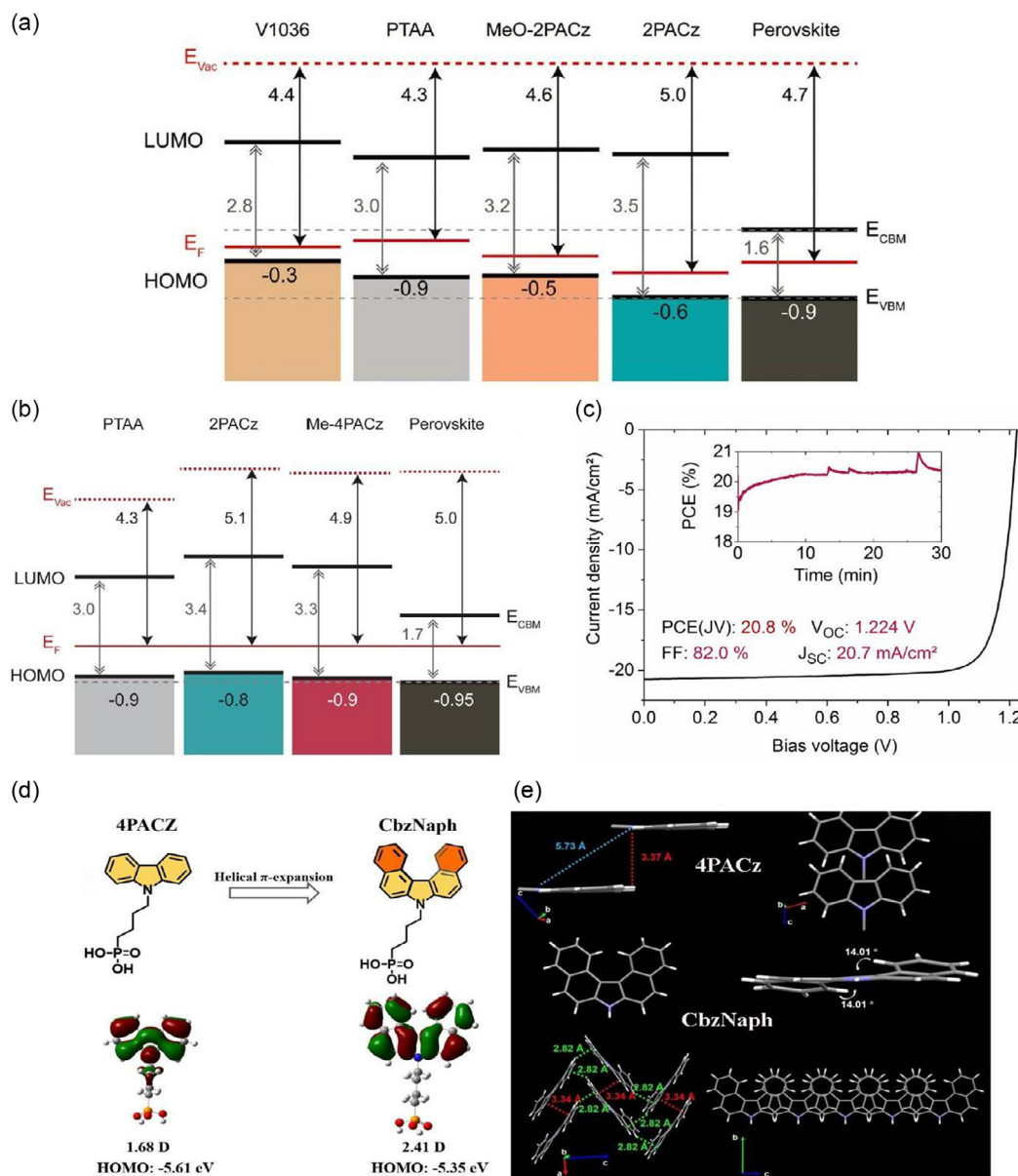


Figure 2. a) Schematic representation of energy alignment between perovskite and PTAA and HSSAMs. Reproduced with permission.^[44] Copyright 2019, The Royal Society of Chemistry. b) Energetic diagram comparing the alignment between perovskite and 2PACz and Me-4PACz HSSAMs and PTAA. c) The current density–voltage (J – V) curves of inverted PSC with Me-4PACz HSSAM. Reproduced with permission.^[46] Copyright 2020, Science. d) Molecular structures, dipole moments, HOMO orbital distributions, and energy levels of 4PACz and CbzNaph. e) The molecular packing patterns corresponding to the π -scaffolds of 4PACz and CbzNaph. Reproduced with permission.^[47] Copyright 2022, Wiley-VCH.

The large dipole moment and compact packing of HSSAMs are favorable to tune the WF of substrate and minimize the leakage current, which can effectively improve the photovoltaic performance of the solar cells. However, the planar and symmetric structure of carbazole moiety usually leads to a small dipole moment of carbazole-based HSSAM. Introducing the polar groups to enhance the molecule dipole often induces the steric hindrance that remarkably undermines the capability in forming a compact HSSAM. Recently, Jen et al. reported a novel design strategy for carbazole-based HSSAM to simultaneously strengthen their dipole moment and promote their dense assembly.^[47] Through the helical π -expansion of carbazole moiety, they synthesized a (4-(7H-dibenzo[c,g]carbazol-7-yl)butyl)phosphonic acid (CbzNaph) HSSAM for inverted PSCs. Compared with the pristine 4PACz ([4-(9H-carbazol-9-yl)butyl]phosphonic acid) HSSAM, CbzNaph had a larger dipole moment and upshifted highest occupied molecular orbital (HOMO) energy level (Figure 2d), which were favorable to modulate the WF of substrate and form a matched energy alignment with perovskite layer.^[43,48,49] Moreover, the helical π -expansion remarkably enhanced the π - π interaction between the carbazole-based groups, which induced the formation of an ordered and compact CbzNaph SAM^[50,51] (Figure 2e). As expected, CbzNaph displayed a faster hole transport and more efficient hole extraction compared with 4PACz. The fabricated inverted PSC with an ITO/CbzNaph HSSAM/perovskite/C₆₀/BCP/Ag structure delivered a PCE of 21.4%, increased by $\approx$ 47% compared to 4PACz device.

Although the carbazole-based SAMs are promising HSSAMs in inverted PSCs, the carbazole block is expensive. The low-cost HSSAM materials are obviously more suitable for large-scale deployment of PSCs. The phenothiazine moiety is a cheap building block and displays high hole mobility and chemical stability.^[52] 2-(3,7-dibromo-10H-phenothiazin-10-yl)ethyl)phosphonic acid (Br-2EPT) is a phenothiazine-based organic molecule containing bromide group, which can self-assemble on the surface of FTO substrate to form a HSSAM. Br-2EPT HSSAM could enhance the adhesion between the perovskite film and the substrate and passivate the interface defects via Lewis acid–base interaction.^[53–55] Furthermore, compared with the commonly used PTAA HTL, Br-2EPT HSSAM showed a lower energy loss for the hole extraction from perovskite layer and larger energy barrier for the electron transport, demonstrating a superior hole selectivity. The fabricated inverted PSC with a FTO/Br-2EPT HSSAM/Cs_{0.05}(FA_{0.92}MA_{0.08})_{0.95}Pb(I_{0.92}Br_{0.08})₃/C₆₀/BCP/Cu structure delivered a PCE of 22.44%, which is much higher than that of the device using PTAA HTL.^[56] Hong et al. found that the sulfur atom in the phenothiazine moiety plays an important role in determining the performance of phenothiazine-based HSSAMs. By replacing S with Se or O in the phenothiazine group, they synthesized two phenothiazine analogs, 2-(3,7-dibromo-10H-phenoselenazine-10-yl)ethyl)phosphonin acid (Br-2EPSe) and 2-(3,7-dibromo-10H-phenoxazine-10-yl)ethyl)phosphonin acid (Br-2EPO) (Figure 3a).^[57] All of Br-2EPSe, Br-2EPT, and Br-2EPO could self-assemble on the surface of FTO substrate to form SAMs with strong dipole moment, leading to the increase in the WF of SAM-coated FTO substrate (Figure 3b). The interface energy level diagrams of FTO/perovskite, FTO/Br-2EPSe/perovskite, FTO/Br-

2EPT/perovskite, and FTO/Br-2EPO/perovskite (Figure 3c) demonstrated that the introduction of Br-2EPSe SAM resulted in the most optimized interface energy alignment for efficiently extracting the hole and blocking the electron from perovskite. In addition, the interaction between the perovskite and the investigated SAM molecules strengthened in the order of Br-2EPO < Br-2EPT < Br-2EPSe. The strong interaction is in favor of reducing interface trap density and enhancing the carrier lifetime. Consequently, the PCE of the inverted PSC with Br-2EPSe HSSAM is higher than those of the device with Br-2EPT and Br-2EPO HSSAMs (Figure 3d). In addition, the unencapsulated device with Br-2EPSe HSSAM preserved $\approx$ 96% of its initial PCE after 500 h of maximum power point (MPP) tracking test under ambient condition (Figure 3e), suggesting an excellent stability.

The triphenylamine-based derivatives are also the ideal head groups of HSSAM molecules due to their superior hole-transporting and electron-donating properties.^[58,59] Introducing the methoxyl groups at the para- and ortho-positions is in favor of molecule overlap and forming the ordered and compact triphenylamine-based HSSAMs.^[60] 4'-[bis(2',4'-dimethoxybiphenyl-4-yl)amino]-biphenyl-4-carboxylic acid (MC-43) molecule (Figure 4a) could directly assemble at the surface of ITO to form a dense HSSAM, which ensured the homogeneous deposition of perovskite film. The HOMO level of MC-43 was determined to be -5.1 eV, which is well aligned with the energy level of perovskite layer (Figure 4b), leading to the fast hole extraction from perovskite and the reduction of the charge losses in the device. Consequently, the fabricated inverted MAPbI₃ PSC with MC-43 HSSAM achieved a PCE of 17.3%.^[33] To improve the performance of triphenylamine-based HSSAMs, the donor–acceptor (D–A) structure groups using the triphenylamine-based moiety as the donor unit were designed and employed to construct HSSAM molecules. The D–A structure had been used for designing dopant-free small-molecule HTL of PSCs and proved to be beneficial for enhancing the hole extraction and reducing charge recombination in PSCs.^[61–65] Zeng et al. designed a D–A structural self-assembled molecule (PPA) (Figure 4c) with triphenylamine donor unit, benzo[c][1,2,5]thiadiazol acceptor unit, and phosphonic acid anchoring group.^[66] Electrostatic potential distribution analysis indicated that the electron-rich area was mainly distributed on the phosphonic acid group (Figure 4c), leading to the strong interaction between PPA HSSAM with both ITO substrate and perovskite layer. As shown in Figure 4d, PPA molecules preferred to pack on the surface of ITO substrate with a face-on orientation due to their planar and rigid structure. The HOMO and lowest unoccupied molecular orbital (LUMO) energy levels of PPA molecule are -5.2 and -2.82 eV, respectively, which are well matched with the energy level of perovskite layer. These favorable features enhanced the hole extraction from perovskite and remarkably reduced the charge recombination. Consequently, the fabricated inverted PSC with an ITO/PPA HSSAM/perovskite/PCBM/BCP/Ag structure achieved a high PCE of 23.24%. In addition, the unencapsulated inverted PSC with PPA HSSAM showed an excellent operational stability, remaining 91% of the initial PCE after 1000 h of MPP tracking. These results highlighted the great superiority of the D–A type HSSAM in promoting the performance of inverted PSCs.

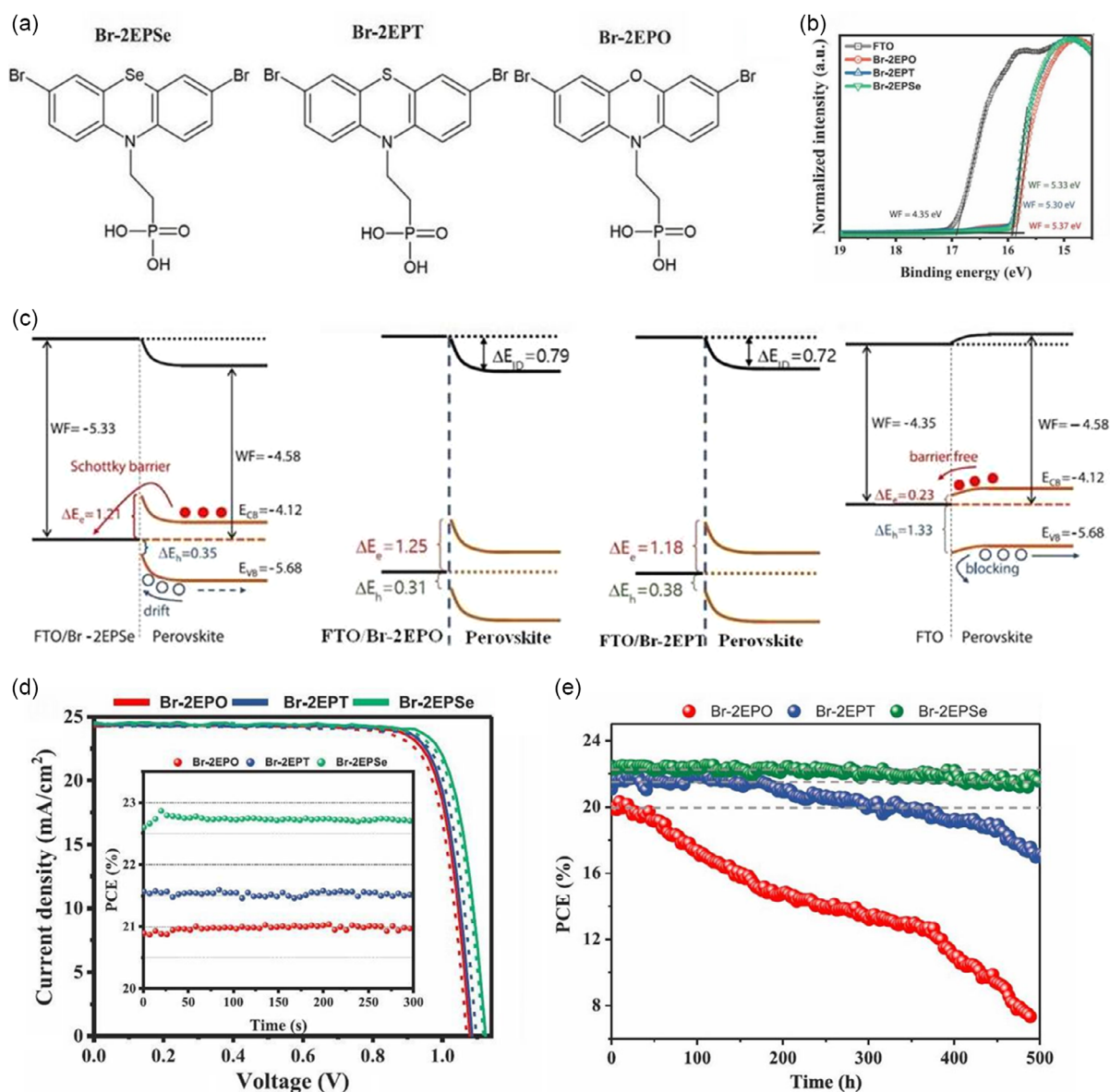


Figure 3. a) Molecular structures of Br-2EPSe, Br-2EPO, and Br-2EPT. b) The secondary electron cutoff region of FTO, FTO/Br-2EPSe, FTO/Br-2EPO, and FTO/Br-2EPT. c) The energy level diagram of FTO/perovskite and FTO/HSSAMs/perovskite interfaces. d) J–V curves of inverted PSCs with Br-2EPSe, Br-2EPO, and Br-2EPT HSSAMs. e) The long-term stability of the unencapsulated device with Br-2EPSe, Br-2EPO, and Br-2EPT HSSAMs under MPP continuous load. Reproduced with permission.^[57] Copyright 2022, Wiley-VCH.

3.1.2. The Anchoring Groups of HSSAMs

The anchoring group is responsible for connecting HSSAM molecules to the surface of TCO substrates through the chemical attachment. Therefore, the anchoring group can influence the dipole moment of HSSAM molecules, the binding strength of HSSAM molecules with TCO substrate, and the coverage of HSSAMs, and thus has a considerable effect on the photovoltaic performance of assembled inverted PSCs. Wu et al. synthesized a series of phenothiazine-based molecules (TPT-S6, TPT-P6, TPT-C6, and TPT-H6) with different anchoring groups (i.e., –COOH, –SO₃H, and –PO₃H₂ groups) (Figure 5a) to form HSSAMs in

inverted PSCs and systematically investigate their performance.^[67] The theoretical calculation indicated that the molecule with –PO₃H₂ group tended to be chemisorbed on the ITO substrate via the tridentate mode while the molecules with –COOH and –SO₃H groups were prone to be adsorbed via the bidentate chemisorption mode (Figure 5b). The binding strength between the –PO₃H₂ group and the ITO substrate was stronger than those between the –COOH and –SO₃H groups and the ITO substrate. Therefore, the TPT-P6 with –PO₃H₂ displayed a larger adsorption amount on the ITO substrate compared to the TPT-C6 with –COOH and TPT-S6 with –SO₃H, leading to forming a more dense and ordered HSSAM. The dense and ordered HSSAM

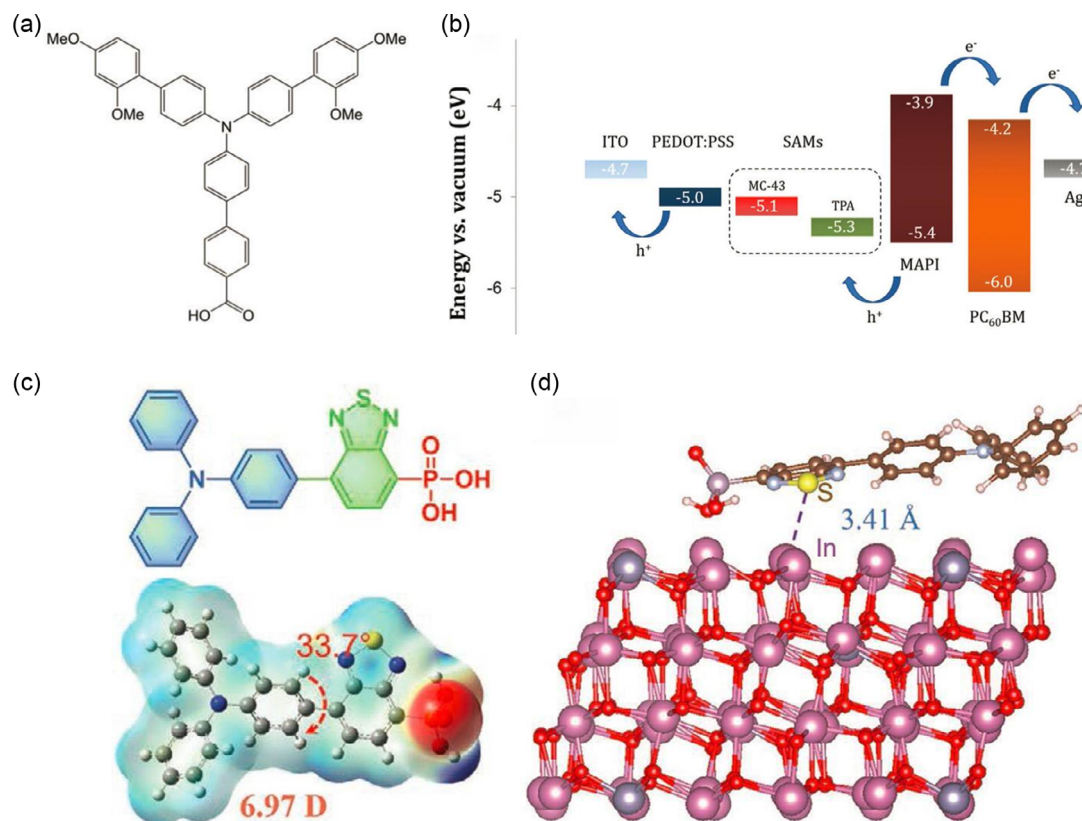


Figure 4. a) The molecular structure of MC-43. b) The energy level diagram of inverted PSC with MC-43 HSSAM. Reproduced with permission.^[34] Copyright 2019, The Royal Society of Chemistry. c) The molecular structure and electrostatic potential surface of PPA. d) Packing model of PPA molecule on the surface of ITO substrate. Reproduced with permission.^[66] Copyright 2023, Wiley-VCH.

is in favor of the efficient hole extraction and transport and the suppression of charge recombination at the interface. Consequently, the fabricated inverted PSC with TPT-6P HSSAM delivered a largely improved PCE of 21.43% compared to the device with TPT-C6 and TPT-S6 HSSAMs. Guo et al. synthesized three D–A type HSSAM molecules (MPA-BT-CA, MPA-BT-BA, MPA-BT-RA) with different anchoring groups (benzoic acid (BA), 2-cyanoacrylic acid (CA), and rhodanine-3-propionic acid (RA)) (Figure 5c).^[68] They found that MPA-BT-CA and MPA-BT-BA molecules vertically self-assembled on the ITO substrate to form the homogeneous HSSAM with smooth surface while MPA-BT-RA molecules tended to obliquely aggregate on the ITO substrate to form multilayer film (Figure 5c). The MPA-BT-CA HSSAM is denser than MPA-BT-BA HSSAM. The dense and homogeneous monolayer with smooth surface is favorable to forming perovskite film with large grains and low defect density. In addition, the dipole moment of MPA-BT-CA is larger than those of MPA-BT-BA and MPA-BT-RA because the electron-withdrawing ability of CA group is stronger than that of BA and RA groups. Therefore, MPA-BT-CA HSSAM could more efficiently extract and transport hole. In addition, the -CN of CA group could efficiently passivate the perovskite defects.^[69] Consequently, the inverted PSCs with MPA-BT-CA HSSAM achieved a PCE of 21.81%, which was much higher than the efficiency of devices with MPA-BT-BA and MCA-BT-RA.

Based on the early discussion, it can be found that the Brønsted–Lowry acids were the most commonly used anchoring groups for HSSAM molecules, such as phosphonic acid, carboxylic acid, and sulfonic acid. Nonetheless, the TCO (ITO and FTO) substrates are sensitive to the strong acidic environment, which has a negative effect on the long-term stability of the fabricated devices.^[70,71] Therefore, the anchoring groups with the weak acidity would be more preferable to HSSAM in inverted PSCs. Boric acid is a weak acid and can alleviate the corrosion of TCO substrate. Importantly, boric acid group can rapidly adsorb on the surface of the TCO substrates through an exothermal process of chemisorption. These features make the boric acid a promising anchoring group for HSSAM in inverted PSCs. Wu et al. designed a triphenylamine-based molecule with boric acid anchoring group (MTPA-BA). MTPA-BA molecules could adsorb on the surface of ITO substrate via the strong interaction between the boric acid group and ITO substrate to form a dense HSSAM, which enhanced the affinity of the substrate for perovskite precursor solution, resulting in a compact contact between the substrate and deposited perovskite layer. In addition, the boric acid group is preferentially adsorbed at the oxygen vacancy sites of ITO surface, which could enhance the stability of the resulting devices. Consequently, the inverted PSC based on MTPA-BA HSSAM delivered a PCE of 22.62% with a high FF of 85.2%.^[72] Moreover, the device showed a largely enhanced

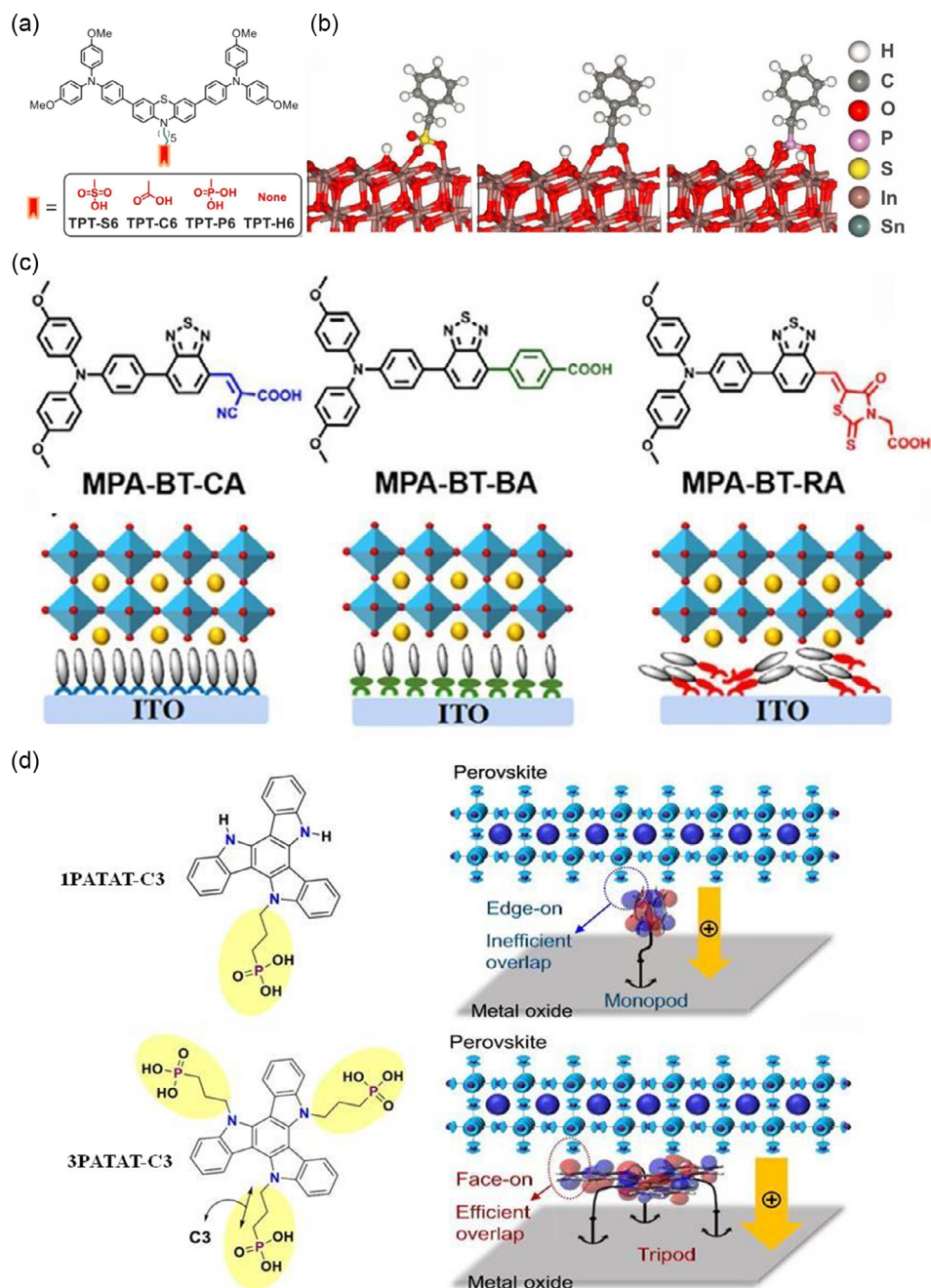


Figure 5. a) Molecular structure of TPT-S6, TPT-C6, and TPT-P6. b) Adsorption mode of TPT-S6, TPT-C6, and TPT-P6 on the surface of ITO substrate. Reproduced with permission.^[67] Copyright 2022, Wiley-VCH. c) Molecular structure and self-assembled mode of MPA-BT-CA, MPA-BT-BA, and MPA-BT-RA. Reproduced with permission.^[68] Copyright 2022, Elsevier. d) Molecular structure of 1PATAT-C3 and 3PATAT-C3 and schematic edge-on and face-on molecular orientation of 1PATAT-C3 and 3PATAT-C3 on ITO substrate. Reproduced with permission.^[73] Copyright 2022, The American Chemical Society.

stability with a preservation of 90% of initial PCE after 2400 h of storage.

To promote the hole extraction efficiency of HSSAM, a novel multipodal strategy for realizing the face-on orientation of HSSAM molecules with respect to TCO substrate was proposed by Truong et al.^[73] They designed and synthesized a series of triazatruxene-based molecule with multiple phosphonic acid

anchoring groups (Figure 5d). The triazatruxene-based molecule with one phosphoric acid group (1PATAT-C3) tended to vertically align (edge-on) on the substrate surface, while the triazatruxene-based molecule with three phosphoric acid groups (3PATAT-C3) preferred to align face on the TCO surface (Figure 5d). Compared to the edge-on oriented 1PATAT-C3 HSSAM, the face-on configuration of 3PATAT-C3 HSSAM resulted in the

more efficient orbital overlap with perovskite and thus remarkably enhanced the hole extraction and reduced interface recombination. Consequently, the assembled inverted PSC with 3PATAT-C3 HSSAM achieved a high PCE of 23% along with an excellent operational stability. These results demonstrated that the optimization of molecular structure and orientation of HSAAMs should result in the further promotion of the inverted PSC performance.

3.1.3. The Spacer Groups of HSSAMs

The spacer group of HSSAM molecules is the backbone that links the anchoring group and the head group, determining the molecule ordering and the lateral noncovalent interaction. The alkyl chain is the most used spacer group in state-of-the-art HSSAMs.^[33,48] The length of alkyl chain influences the thickness and the charge transport properties of HSSAMs. The long alkyl chain results in the strong Van Der Waals interaction, which favors the vertical alignment of HSSAM molecules on the substrate. However, the increase of alkyl chain length can lead to an increase in the tunneling barrier width, which reduces the hole transfer rate and introduces the additional series resistance into the devices. This dilemma can be circumvented by adopting the conjugated spacer group in HSSAMs.

The conjugated spacer group could modulate the HOMO and LUMO energy levels of HSSAMs, which enhanced the competency of the hole extraction and transport and the electron blocking of HSSAMs. The changing of the conjugation length could manipulate the molecular self-assembly behavior, and the length extension of the conjugated group could favor the formation of a dense HSSAM because of the twisted configuration of the head groups. In addition, the HOMO and LUMO orbitals of HSSAM molecules with alkyl spacer group both locate on the electron-rich moiety while the HOMO and LUMO orbitals of HSSAM molecules with conjugated spacer group are spatially separated and delocalized, as shown in **Figure 6a**. The electron delocalization and frontier energy level modulation induced by the incorporation of the conjugated spacer group greatly enhanced the photostability and electrochemical stability of HSSAMs. To investigate the effectiveness of HSSAMs with the conjugated spacer group, Zhu et al. fabricated inverted PSCs with MeO-2PACz HSSAM with the nonconjugated spacer group and MPA-Ph-CA HSSAM with the conjugated spacer group and analyzed their performance.^[74] The inverted PSC with MeO-2PACz HSSAM achieved a PCE of 21.29%, while the device with MPA-Ph-CA HSSAM delivered a greatly promoted PCE of 22.53% (**Figure 6b**). Furthermore, the inverted PSC with MPA-Ph-CA HSSAM displayed an excellent stability. In the dark with the

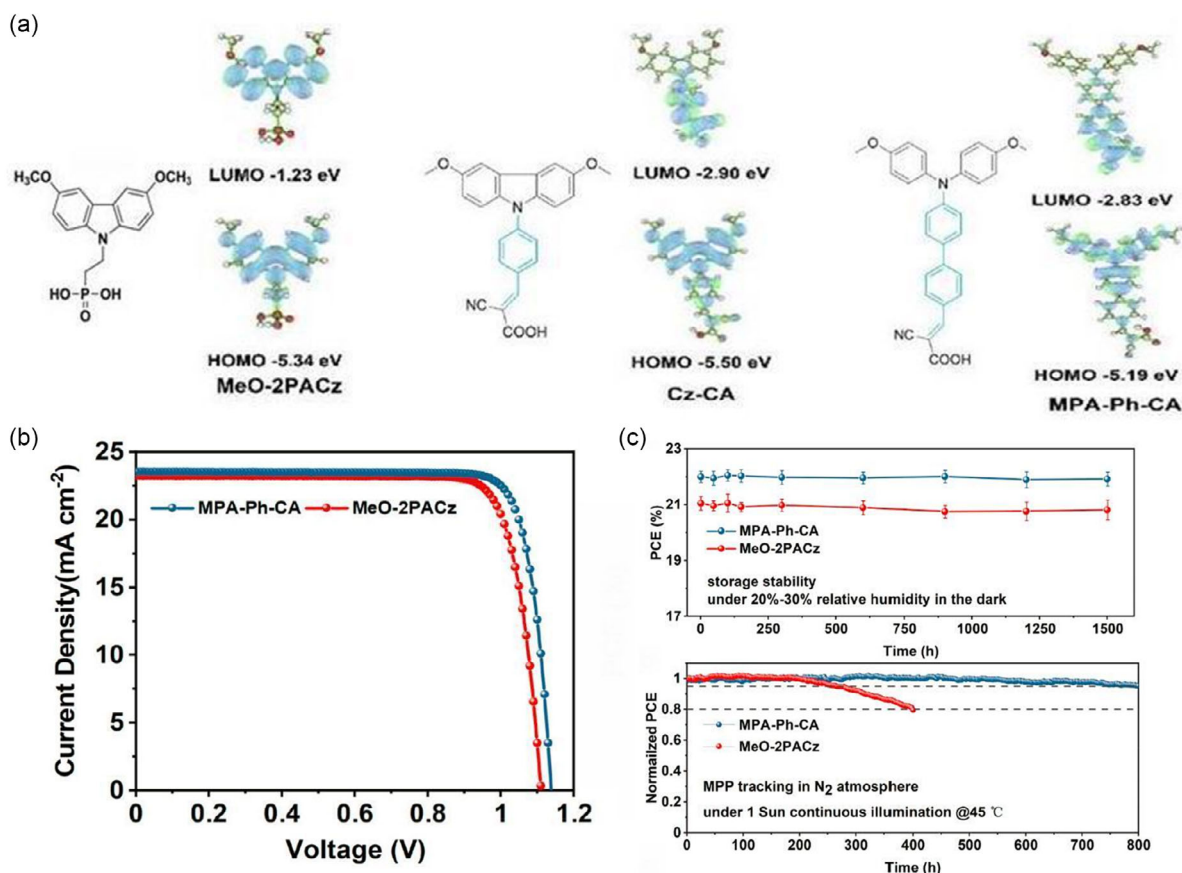


Figure 6. a) Molecular structure and density functional theory-based calculations of MeO-2PACz, Cz-CA, and MPA-Ph-CA HSSAMs. b) *J*-*V* curves of inverted PSCs with MeO-2PACz and MPA-Ph-CA HSSAMs. c) The stability of the device with MPA-Ph-CA HSSAM. Reproduced with permission.^[74] Copyright 2022, The American Chemical Society.

relative humidity of 20–30%, the PCE of the MPA-Ph-CA HSSAM-based device is almost constant. Under continuous illumination in N_2 atmosphere at 45 °C, the inverted PSC with MPA-Ph-CA HSSAM preserved over 95% of its initial PCE after 800 h continuous illumination, while the PCE of the cell with MeO-2PACz HSSAM decreased to 80% of its initial value after 400 h continuous illumination (Figure 6c). These results demonstrated that the HSSAMs with conjugated spacer group are promising for fabricating the efficient and stable inverted PSCs.

3.2. The Synergistic Coassembly of HSSAM for Inverted PSCs

Generally, a dense HSSAM is critical for good performance and high stability of inverted PSCs because it can prevent the direct contact between the perovskite layer and the TCO substrate that can induce the leakage current and serious interface recombination and remarkably deteriorate the device performance. Although the coverage and quality of HSSAM could be greatly improved through the molecule structure design, the deposition of dense and well-ordered HSSAM is still difficult because of the roughness and the inhomogeneous surface chemistry of the TCO substrate, which results in the inevitable presence of defects in HSSAM.^[75,76] Generally, the formation process of the SAM consists of the adsorption of molecules and the rearrangement and reorientation of molecules.^[77] Therefore, improving the HSSAM molecule adsorption must be an effective strategy for forming a dense HSSAM with minimal defects.^[53,78,79] Introducing NiO_x interlayer onto the surface of TCO substrate could modify the TCO surface chemistry and increase the surface hydroxyl density, which could promote the chemisorption of HSSAM molecules.^[80,81] Sun et al. demonstrated that sputtering NiO_x interlayer onto ITO substrate provided more binding site for the chemisorption of MeO-2PACz molecules,^[80] which led to forming tridentate binding of MeO-2PACz molecule on the substrate (Figure 7a). This tridentate binding mode promoted

the chemisorption of MeO-2PACz molecule and then resulted in the formation of a full-coverage MeO-2PACz HSSAM. In addition, the combination of MeO-2PACz HSSAM with NiO_x interlayer optimized the energy level alignment at the TCO/perovskite film interface and led to a good interface property of inverted PSCs (Figure 7b). These features favored the hole extraction and electron blocking. Consequently, the inverted PSC based on MeO-2PACz HSSAM achieved a high PCE of $\approx 20\%$ (Figure 7c).

Introducing the small aliphatic molecules into SAM molecules as the filler is also a promising strategy to promote the quality of the formed monolayer.^[82,83] Incorporating butylphosphonic acid (C4) molecules into V1306 HSSAM improved its coverage through filling the gaps left by large V1306 molecules, decreasing the interface recombination, and enhancing the performance of resulting devices.^[29] 2PACz is a commonly used HSSAM molecule. Owing to the rough surface of TCO substrate and large size of 2PACz molecule, it is difficult to form a perfect 2PACz HSSAM with full coverage on the TCO substrate. The existence of uncovered TCO substrate would result in the direct contact between perovskite layer and TCO substrate and increase the charge recombination and leakage current at the interface. Hayase et al. demonstrated that the introduction of a small methyl phosphonic acid (MPA) molecule through a coadsorption strategy could effectively fill the blank space of 2PACz HSSAM coated on TCO substrate (Figure 8a), leading to the formation of high-quality 2PACz/MPA HSSAM with full coverage and high packing density.^[84] The high-quality 2PACz/MPA HSSAM enabled the deposition of compact $CS_{0.025}FA_{0.475}MA_{0.5}Pb_{0.5}Sn_{0.5}I_{2.925}Br_{0.075}$ perovskite film with high stability, large grains grown in the vertical direction, and decreased trap density (Figure 8b). The assembled Pb–Sn PSC with a FTO/2PACz/MPA HSSAM/perovskite/PCBM/ C_{60} /BCP/Ag structure delivered a high PCE of 23.3% and showed an excellent long-term stability.

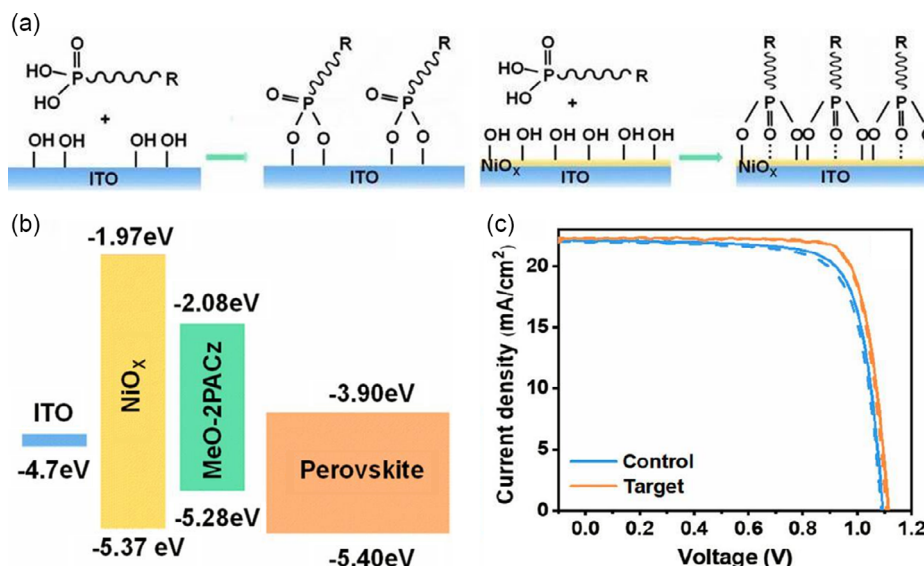


Figure 7. a) The schematic diagrams of bidentate and tridentate bonding of MeO-2PACz on ITO and ITO/ NiO_x , respectively. b) The energy level diagrams of MeO-2PACz HSSAM at the perovskite/ITO interface. c) J - V curves of MeO-2PACz HSSAM-based inverted PSC with (target) and without (control) NiO_x . Reproduced with permission.^[80] Copyright 2021, Wiley-VCH.

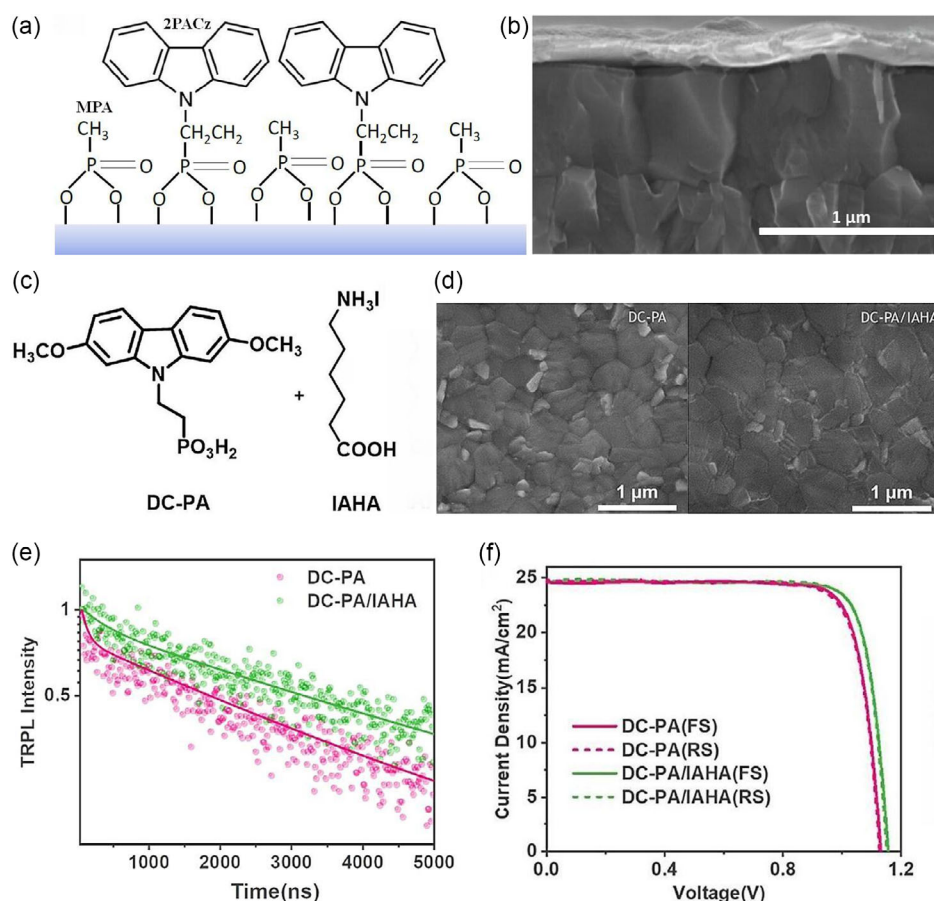


Figure 8. a) Schematic showing the formation of 2PACz/MPA co-HSSAM on the substrate. b) Cross-sectional SEM image of inverted PSC with 2PACz/MPA HSSAM. Reproduced with permission.^[84] Copyright 2022, The American Chemical Society. c) The molecular structures of DC-PA and IAHA. d) SEM images of perovskite films deposited on DC-PA and DC-PA/IAHA HSSAMs. e) Time-resolved photoluminescence of perovskite films deposited on DC-PA and DC-PA/IAHA HSSAMs. f) J–V curves of inverted PSCs with DC-PA and DC-PA/IAHA HSSAMs. Reproduced with permission.^[85] Copyright 2022, Wiley-VCH.

To minimize the defect and promote the quality of ((2,7-dimethoxy-9H-carbazol-9-yl) methyl) phosphonic acid (DC-PA) HSSAM, Jen et al. incorporated 6-(iodo-λ⁵-azanyl) hexanoic acid (IAHA) molecules into DC-PA HSSAM as the second component (Figure 8c).^[85] Through delicately tuning the ratio of DC-PA to IAHA, a coassembled HSSAM with dense coverage was formed on the ITO substrate. This favored the deposition of high-quality perovskite film (Figure 8d) and prevented the direct contact between the substrate and perovskite layer, leading to the decrease of the leakage current in assembled device. Furthermore, the ammonium group of IAHA could coordinate with the perovskite, passivating the interface defects and suppressing the interface recombination (Figure 8e).^[86,87] Consequently, the inverted PSC with DC-PA/IAHA co-HSSAM delivered a PCE of 23.59%, which is higher than the efficiency of the device with DC-PA HSSAM (22.64%) (Figure 8f). In addition, the co-HSSAM-based inverted PSC showed an excellent stability with maintaining 97% of the initial efficiency after 450 h of storage at 65 °C.

Undoubtedly, introduction of HSSAM is one of the most promising strategies to remarkably improve the performance

and enhance the manufacturability of inverted PSCs. Nevertheless, many commonly used HSSAMs in inverted PSCs, such as Me-4PACz and 2PACz, are hydrophobic with respect to perovskite precursor solution, resulting in the poor coverage and morphology of the obtained perovskite film that deteriorated the photovoltaic performance and stability of the device. To promote the wettability of perovskite precursor solution, the hydrophilic organic molecules as the second component were incorporated into HSSAMs, creating a co-assembly structure with the hydrophilic surface. Introducing 1,6-hexylenediphosphonic acid (6dPA) molecules to Me-4PACz precursor solution resulted in the formation of the coassembly HSSAM on ITO substrate (Figure 9a). The presence of phosphonic acid groups on the HSSAM surface enhanced the surface energy and promoted the surface wettability of HSSAM-covered ITO substrate (Figure 9b), resulting in the deposition of perovskite film with full coverage. The assembled inverted PSC with an ITO/HSSAM/FA_{0.75}Cs_{0.22}MA_{0.03}Pb(I_{0.82}Br_{0.15}Cl_{0.03})₃/LiF/C₆₀/BCP/Ag structure delivered a PCE of 20.9%.^[88] Kabra et al. demonstrated that the introduction of poly(9,9-bis(3-(N,N-dimethyl)-N-ethylammonium-propyl-2,7-fluorene)-alt-2,7-(9,9-dioctylfluorene))

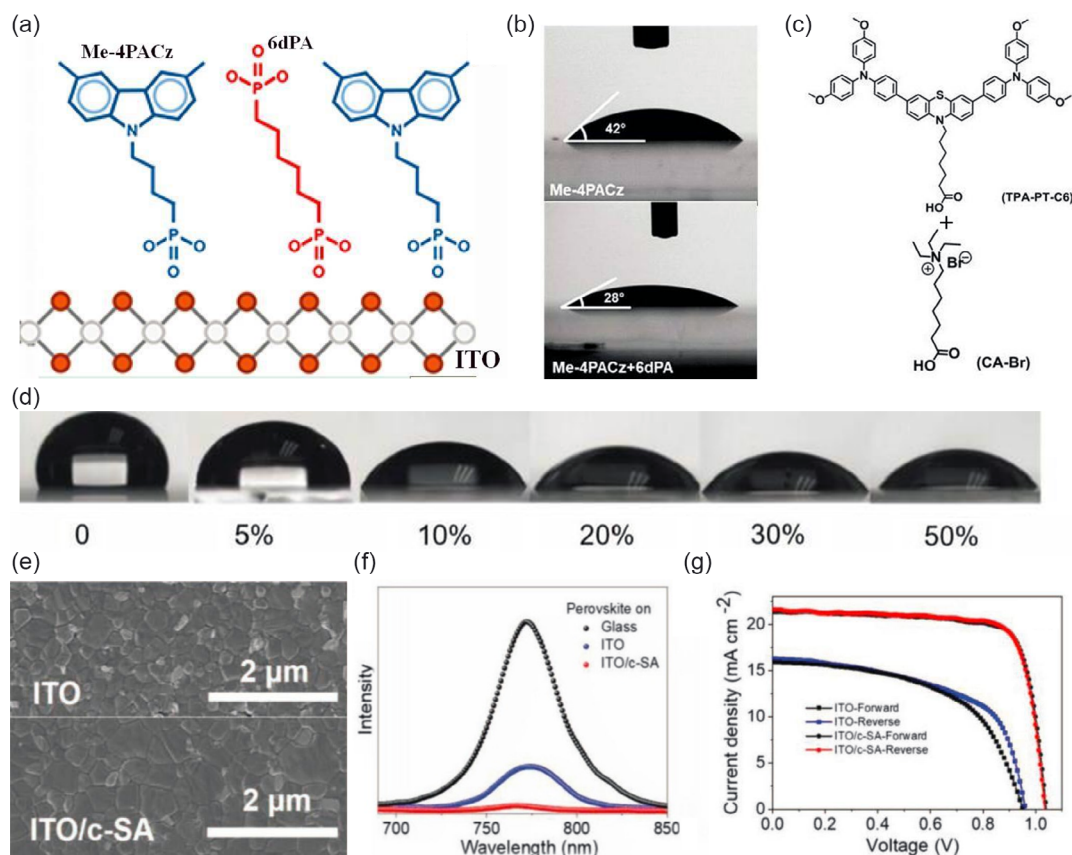


Figure 9. a) Schematic illustration of coassembly of Me-4PACz and 6dPA on ITO substrate. b) The contact angles of perovskite precursor solution on ITO/Me-4PACz and ITO/Me-4PACz/6dPA. Reproduced with permission.^[88] Copyright 2023, The American Chemical Society. c) The molecular structure of TPA-TP-C6 and CA-Br. d) Water contact angle on TPA-TP-C6 HSSAMs with different amounts of CA-Br. e) SEM images of MAPbI₃ perovskite film deposited on ITO and ITO/TPA-TP-C6/CA-Br (ITO/c-SA). f) The steady-state photoluminescence spectra of MAPbI₃ perovskite film on bare glass, ITO, and ITO/c-SA. g) J–V curves of inverted PSCs based on ITO and ITO/c-SA (30% CA-Br). Reproduced with permission.^[90] Copyright 2019, Wiley-VCH.

dibromide (PFN-Br) also reduced the hydrophobicity of Me-4PACz HSSAM. The inverted PSC with Me-4PACz/PFN-Br co-HSSAM achieved a PCE over 20% for a wide-bandgap (1.6 eV) perovskite.^[89]

Zhu et al. synthesized a new HSSAM molecule, 7-(3,7-bis(4-(bis(4-methoxyphenyl)amino)phenyl)-10 H-phenothiazin-10-yl)heptanoic acid (TPA-PT-C6), and employed 6-carboxy-N,N,N-triethylhexan-1-ammonium bromide (CA-Br) as the second component to modified the TPA-PT-C6 HSSAM (Figure 9c).^[90] The introduction of CA-Br with the hydrophilic quaternary ammonium group modulated the surface property of TPA-PT-C6 HSSAM and enhanced the wettability of perovskite precursor on TPA-PT-C6 HSSAM (Figure 9d), which improved the perovskite deposition, ameliorated the perovskite quality (Figure 9e), and enhanced the contact between the perovskite film and HSSAM. Therefore, the TPA-PT-C6/CA-Br coassembly HSSAM displayed a superior hole-extraction capability as evidenced by photoluminescence measurements (Figure 9f). Based on the TPA-PT-C6/CA-Br coassembly HSSAM, the assembled inverted PSC with an area of 1.02 cm² achieved a PCE of 17.9% (Figure 9g) and showed a good stability with 89% of efficiency preserved after 120 days of storage under ambient conditions.

Introducing an overlayer with superwetting capability on top of HSSAM to form a self-assembled bilayer was also an effective strategy for ameliorating the surface wettability, which allowed depositing a compact perovskite film with full coverage. Loi et al. employed (2-(3,6-Dibromo-9H-carbazol-9-yl)ethyl)phosphonic acid (Br-2PACz) molecule to form a HSSAM on ITO substrate, and introduced an overlayer composed of carbazole alkylammonium iodide derivative (4CzNH₃I) on top of Br-2PACz HSSAM.^[91] As a result, a self-assembled bilayer with the surface exposed ammonium group was formed through the π - π interaction of two carbazole moieties of Br-2PACz and 4CzNH₃I molecules (Figure 10a). This self-assembled bilayer displayed a good wettability for perovskite precursor solution and did not compromise the excellent hole extraction and transport ability of Br-2PACz HSSAM. Consequently, the perovskite film deposited on the self-assembled bilayer showed a larger grain size compared to that deposited on Br-2PACz (Figure 10b). The fabricated inverted PSC based on the self-assembled Br-2PACz/4CzNH₃I bilayer exhibited a higher efficiency, lower leakage current, and better stability compared with the device based on Br-2PACz HSSAM. The self-assembled bilayer could also be formed by an amphiphilic molecule,

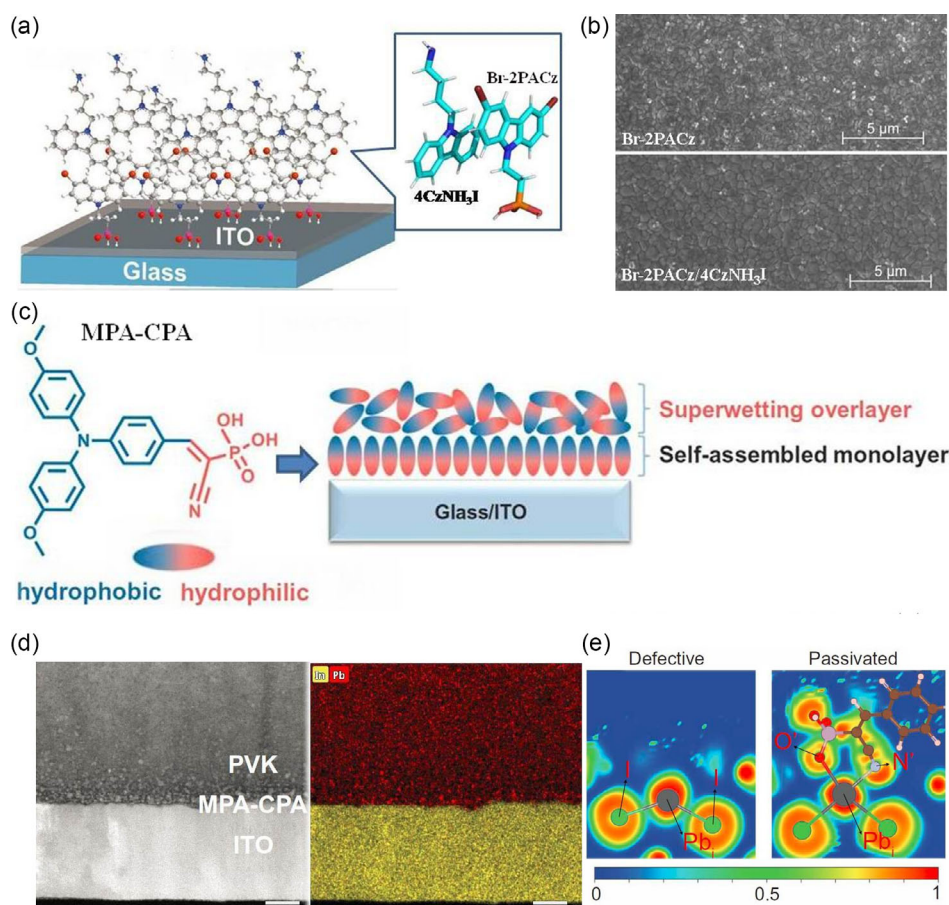


Figure 10. a) Schematic diagram of coassembly of Br-2PACz and 4CzNH₃I. b) SEM images of perovskite film deposited on Br-2PACz and coassembled Br-2PACz/4CzNH₃I. Reproduced with permission.^[91] Copyright 2023, Wiley-VCH. c) Molecular structure of MPA-CPA and schematic diagram of the bilayer stack of amphiphilic MPA-CPA. d) Cross-sectional high-angle annular dark field (HAADF) image and corresponding energy dispersive X-ray spectroscopy (EDX) mapping of the perovskite/MPA-CPA/ITO stack. e) The calculated electron localization function in the region of the defective molecular configuration and the passivated molecular configuration. Reproduced with permission.^[30] Copyright 2023, Science.

(2-(4-(bis(4-methoxyphenyl)amino)phenyl)-1-cyanovinyl)phosphonic acid (MPA-CPA). Spin-coating the MPA-CPA solution onto ITO substrate led to the formation of a bilayer consisted of a chemically adsorbed MPA-CPA HSSAM and an unadsorbed overlayer (Figure 10c). The unadsorbed MPA-CPA overlayer showed a superwettability for perovskite precursor solution, which is favorable for the deposition of highly uniform perovskite film with full coverage, in particular for the deposition of large-area perovskite film. Furthermore, the MPA-CPA overlayer improved the interface contact between the MPA-CPA HSSAM and perovskite layer and passivated the defects in the buried interface region (Figure 10d, e), resulting in the remarkable reduction of the charge recombination at the buried interface. The assembled inverted PSC delivered a certified PCE of 25.4% with a high FF of 84.7%. In addition, 1 cm² device and 10 cm² module achieved PCEs of 23.4% and 22%, respectively.^[30] These results clearly demonstrate that the self-assembled bilayer design is a promising strategy for enhancing the photovoltaic performance of inverted PSCs and promoting the fabrication of large-area inverted PSCs.

4. Deposition Methods of HSSAM in Inverted PSCs

The liquid phase deposition is the most common route used to form HSSAM on TCO substrate for inverted PSCs, including dip- and spin-coating methods. The dip coating is a simple and low-energy process to form HSSAM through the molecular self-assembly, which implies that the TCO substrates are immersed into the HSSAM molecule solution where molecules are chemically adsorbed. The molecule adsorption process can be modulated through controlling the solvent, the concentration, and the dipping time and temperature. Generally, the dip-coating process would take a long time to form HSSAM on the TCO substrate due to the slow molecule absorption. In contrast, the spin coating is a quick and highly controllable route to deposit thin film. It only takes tens of seconds to deposit HSSAM molecules and then several minutes to anneal them. The quality of HSSAM fabricated by a spin-coating method can be optimized through controlling the solution concentration and the spin coating speed and time. Due to its rapid and convenient deposition, the spin coating route is more preferable to the deposition of HSSAM for inverted PSCs.

Generally, the HSSAM molecules are bonded to the surface of TCO substrate through the condensation reaction between the anchoring groups of HSSAM molecule and the surface groups of TCO substrate.^[77,92,93] Therefore, the surface chemistry of TCO substrate, such as the reactivity and concentration of the surface hydroxyl group, must be a key element for modulating the binding strength, surface coverage, and the density of HSSAM.^[93,94] It has been reported that the presence of hydroxyl-group-rich NiO interlayer greatly enhanced the adsorption of MeO-2PACz molecule on ITO substrate and improved the homogeneity and surface coverage of MeO-2PACz HSSAM.^[81] Recently, Wu et al. employed the HF acid and the subsequent UV-O₃ treatments to reconstruct the surface of ITO substrate by removing the terminal hydroxyl group (-OH_T) and the hydrolysis product (such as In(OH)₃), resulting in the increase of the surface activity of ITO substrate. This facilitated the MeO-2PACz molecule adsorption and led to the formation of the dense MeO-2PACz HSSAM on ITO substrate.^[95] As a result, a high-quality perovskite film with large grains and high crystallinity was formed on the MeO-2PACz HSSAM-coated ITO substrate. The dense HSSAM and the high-quality perovskite film synergistically enhanced the hole extraction and decreased the interface recombination. Consequently, the resulted inverted PSC with an ITO/MeO-2PACz HSSAM/perovskite/C₆₀/BCP/Ag structure delivered a remarkably improved efficiency and stability compared with the control device.

It should be noted that most of HSSAM molecules are amphiphilic in nature. The amphiphilic molecules tend to form the micelle in the solution when their concentration exceeds the critical micelle concentration (CMC). The commonly used processing concentration of HSSAM molecules is usually much higher than their CMC. Therefore, the amphiphilic HSSAM molecules are mainly in the form of micelle in the solution, which could affect the self-assembly of molecules on the substrate and result in the formation of the defect in HSSAM. To address this problem, Jen et al. developed a cosolvent strategy to decompose the micelle of HSSAM molecules in the solution.^[96] N,N-dimethylformamide (DMF) was introduced into isopropanol (IPA) solution of the carbazole-based HSSAM molecules, 2PACz, MeO-2PACz, and CbzNaph, to form cosolvent solution. The strong interaction between DMF and the carbazole group effectively decomposed the micelles formed in IPA solution. Consequently, the dense HSSAM with minimal defects on ITO substrate was formed through a spin-coating process using DMF/IPA cosolvent solution (Figure 11a). The dense HSSAM favored the deposition of uniform perovskite film and enabled the matched energy level alignment at the perovskite/HSSAM interface, enhancing the hole extraction and suppressing the recombination. As a result, the inverted PSC with the cosolvent CbzNaph HSSAM delivered a high PCE of 24.02% with a FF of 83.32%. The results reported by Jen et al. can facilitate the development of liquid-phase-deposition HSSAM for more efficient inverted PSCs.

To simultaneously simplify the device fabrication process and circumvent the wetting issue encountered when depositing perovskite layer onto HSSAM, Luther et al. reported a novel one-step spin-coating strategy which enabled the spontaneous formation of HSSAM and perovskite film on ITO substrate.^[97] They incorporated HSSAM molecules into the perovskite

precursor solution and spin-coated it onto ITO substrate. During the perovskite crystallization process, as shown in Figure 11b, HSSAM molecules were pushed and concentrated on the surface of ITO substrate to form a robust HSSAM. This novel codeposition method enabled the simultaneous formation of a high-quality perovskite film with large grains and an excellent HSSAM, which greatly enhanced the hole extraction and reduced the interface recombination. The fabricated inverted PSC achieved a high PCE of 24.5% (Figure 11c) and showed an excellent stability with remaining >90% of the initial efficiency after 1200 h of operation under continuous illumination. This codeposition strategy was compatible with different solvent, HSSAM molecules, and perovskite, and therefore could remarkably improve the manufacturability of inverted PSC (Figure 11d).

Spray coating is also a liquid phase method for depositing high-quality thin films.^[98] However, depositing SAM through a spray coating process was reported minimally in the previous literatures. This may be attributed to that the spray coating process is prone to an excess deposition of self-assembled molecules, impeding the formation of a dense and ordered monolayer. Nevertheless, spray coating permits the deposition over rough surfaces and is performed using a cheap and simple system with the roll-to-roll process capability. This makes the spray coating method a promising strategy to deposit HSSAM for inverted PSCs. Lidzey et al. developed a spray coating process with the preannealing and postannealing rinse protocols to deposit MeO-2PACz HSSAM on ITO substrate (Figure 12a). The combination of the preannealing rinse and postannealing rinse enables the effective removal of the excessive MeO-2PACz molecules and the formation of the dense and ordered MeO-2PACz HSSAM on ITO substrate. The assembled inverted PSC with MAPbI₃ perovskite film deposited by a gas-assisted spray process achieved a PCE of 20.8%.^[99] Lidzey's work reveals the promising of spray coating as a scalable deposition technique for depositing dense HSSAM in inverted PSCs.

In addition to liquid-phase deposition techniques, the vapor-phase deposition was also adopted to deposit HSSAM for inverted PSCs. The vapor-phase deposition method does not require the solvent and therefore is especially fit to HSSAM molecule that has low solubility in solvents. Furthermore, the vapor-phase deposition allows the precise control of the deposition rate and facilitates the uniform and conformal deposition. Therefore, it is a popular technique to deposit the organic functional layer for optoelectronic devices.^[100] Farag et al. reported that the high-quality HSSAMs with the lossless interface properties could be formed on ITO substrate through the vapor-phase deposition process using the most common 2PACz, MeO-2PACz, and Me-4PACz.^[101] They demonstrated that the vapor-phase-processed HSSAMs showed an improved surface wettability compared to the liquid-phase-processed counterparts (Figure 12b), which facilitated the deposition of perovskite precursor solution and promoted the formation of high-quality perovskite film with full coverage. Consequently, the performance of inverted PSC with vapor-phase-processed 2PACz HSSAM was slightly superior to that of the device with liquid-phase-processed 2PACz HSSAM (Figure 12c). Furthermore, the vapor-phase deposition enabled the deposition of high-quality conformal HSSAM on micrometer-sized textured surface, opening a new road for

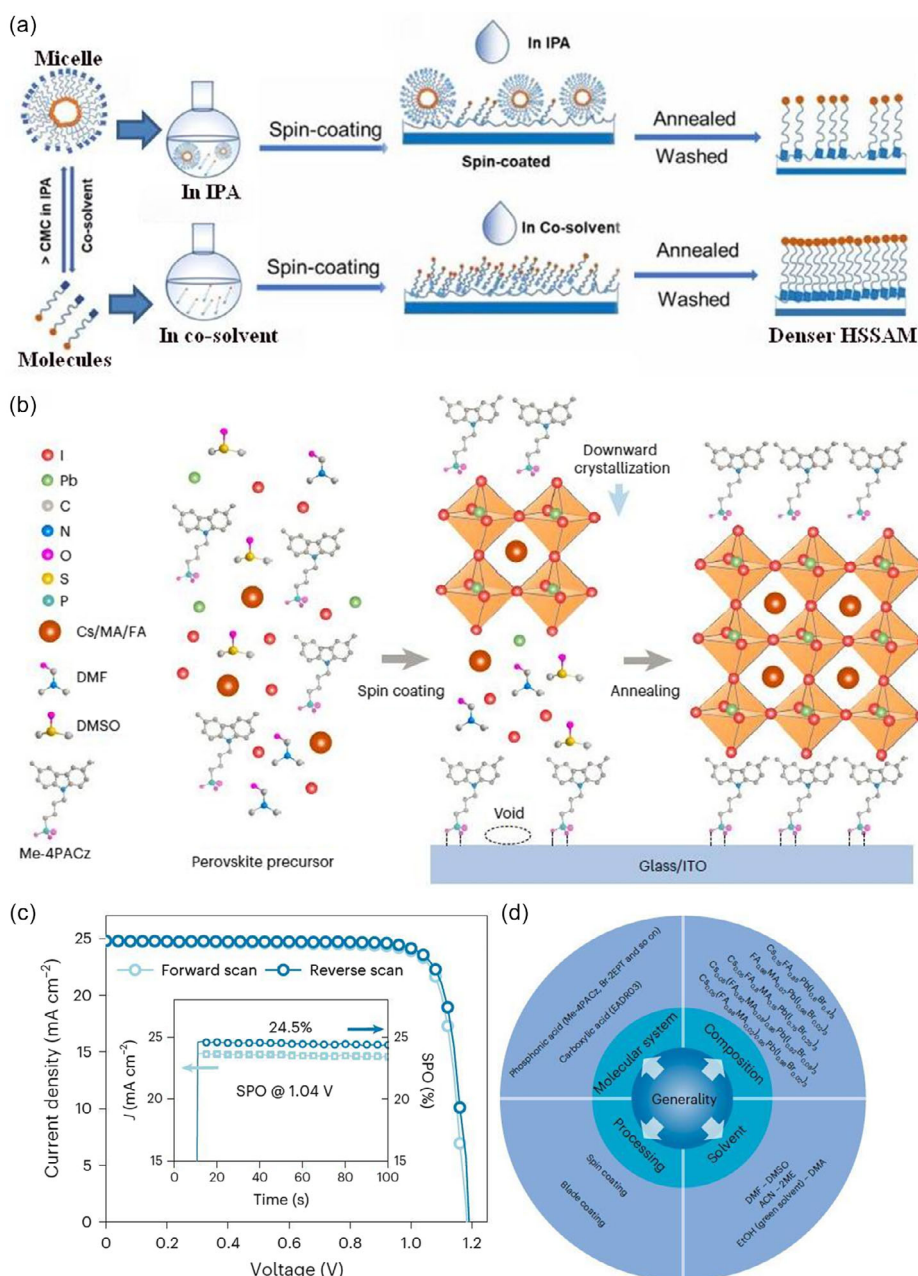


Figure 11. a) Schematic diagram of HSSAM deposition from IPA and cosolvent solution. Reproduced with permission.^[96] Copyright 2023, Wiley-VCH. b) Schematic illumination of HSSAM formation and perovskite crystallization process through a one-step codeposition process. c) *J*-*V* curves of inverted PSC based on Me-4PACz HSSAM and Cs_{0.05}(FA_{0.98}MA_{0.02})_{0.95}Pb(I_{0.98}Br_{0.02})₃ perovskite film deposited by a codeposition process. The inset displayed the stabilized power output. d) Summary of the generality of HSSAM and perovskite codeposition. Reproduced with permission.^[97] Copyright 2023, Nature.

developing perovskite/Si tandem solar cell using the textured Si bottom cell.

5. Conclusion and Perspectives

Since the first attempt in 2018 by Getautis et al., HSSAM strategy has been positioned as a promising method to promote the performance and stability of inverted PSCs. In this review, we have

summarized the development of HSSAMs applied in inverted PSCs. Compared with the conventional HTL used in inverted PSCs, such as PTAA, PEDOT:PSS, and NiO_x, HSSAMs provide numerous advantages that are summarized as follows: 1) the thickness of HSSAM is very small (≈ 1 nm), which offers the benefits of minimal material consumption and low resistance loss and parasitic absorption. 2) The head groups of HSSAMs can interact with perovskite layer, passivating the interface defects and ameliorating perovskite quality. 3) HSSAMs are ordered

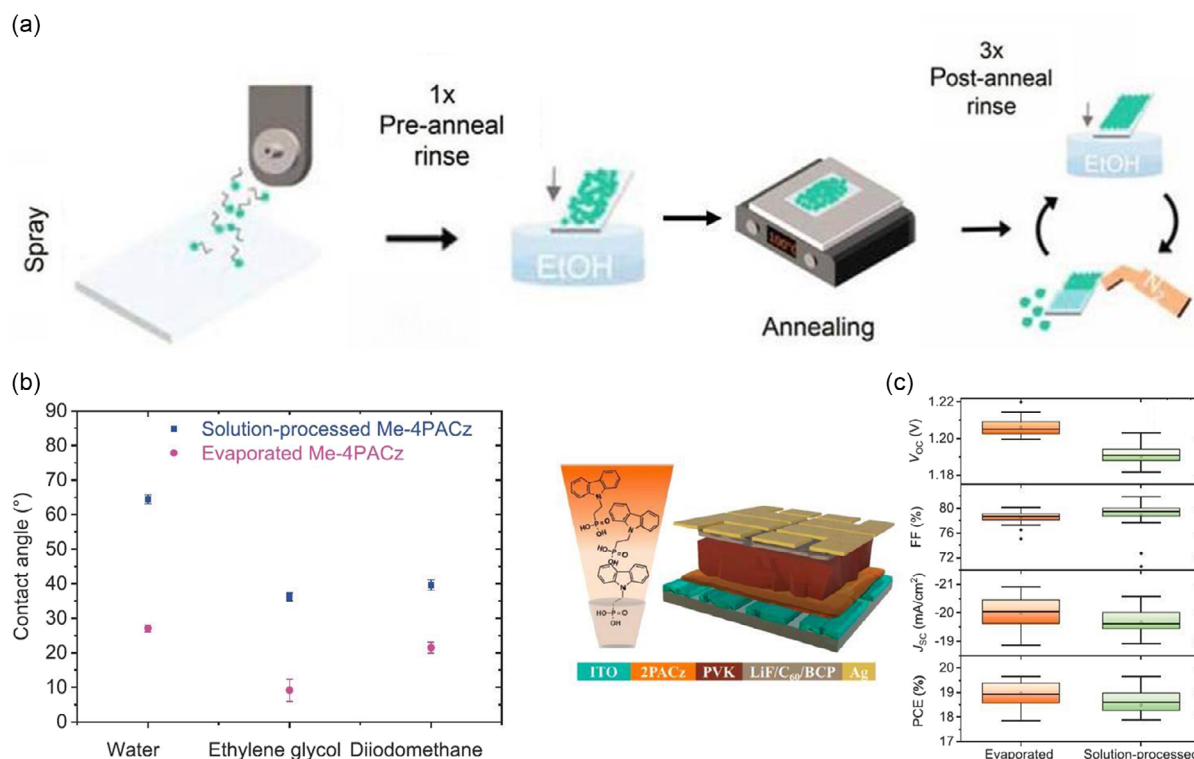


Figure 12. a) Schematic diagram of spray coating process of MeO-2PACz HSSAM. Reproduced with permission.^[99] Copyright 2022, Wiley-VCH. b) The contact of water, ethylene glycol, and diiodomethane droplets on evaporated and solution-processed Me-4PACz HSSAMs. c) The structure and photovoltaic parameters of inverted PSCs based on evaporated and solution-processed 2PACz HSSAMs. Reproduced with permission.^[101] Copyright 2023, Wiley-VCH.

arrays of organic molecules adsorbed on TCO surface. The ordered alignment of HSSAM molecules enables HSSAM to show permanent dipole moment, which contributes to tuning the WF of TCO substrate and forming the matched interface energy-level alignment to enhance the hole extraction and block the electron. 4) HSSAM molecules are connected to the TCO substrate by forming the covalent binding between their anchoring group (such as phosphoric acid, carboxylic acid, and boronic acid) and the hydroxyl groups on the TCO surface. This enables the conformal coverage of HSSAMs, which makes HSSAMs favorable for constructing tandem solar cells that contain perovskite layer. 5) HSSAMs can be deposited onto TCO substrate through different deposition techniques, which is compatible with low cost and scalable production. Benefiting from the advantages of HSSAM, as shown in Table 1, the performance of inverted PSCs based on HSSAMs had been remarkably improved.

In addition to extracting holes from perovskite layer and transporting them to TCO substrate, HSSAM could also affect the morphology and crystallinity of perovskite layer deposited atop it. Therefore, the surface properties of HSSAM are the key factors to fabricate high-quality perovskite layer for inverted PSCs. Generally, the hydrophobic surface could reduce the nucleation rate of perovskite and lead to forming the perovskite layer with large grain size and high crystallinity, while the hydrophilic surface could promote the perovskite nucleation and result in forming compact perovskite layer with full coverage. Therefore,

deliberately modulating the surface wettability of HSSAM through ingeniously designing HSSAM molecular structure and/or adopting the coassembly strategy is of paramount importance for forming high-quality perovskite film and enhancing the performance of inverted PSCs.

Generally, the dense and ordered HSSAMs without defect are favorable for fabricating high-performance inverted PSCs. To promote the quality of HSSAM used in inverted PSCs, the most researchers focused on the HSSAM molecule design and the coating process optimization. In reality, the steps, grain boundaries, and impurities on TCO substrates also induce the formation of defects in HSSAM. However, the effect of the surface properties and defects of TCO substrates on the HSSAM quality seems to be underestimated. Recently, Wu et al. demonstrated that the reconstruction of ITO surface by HF acid and UV-O₃ treatments facilitated the HSSAM molecule adsorption and reduced the HSSAM defects.^[96] Therefore, the systematic investigation on the effect of surface characteristics of TCO substrate on the HSSAM formation is needed. It is expected that the surface treatment and modification of TCO substrate must promote the deposition of high-quality HSSAM.

Although numerous molecular structures have been reported for HSSAM that constructed high-performance inverted PSCs, designing novel molecular structures for HSSAM is still highly desirable to simultaneously passivate the interface defects, tune the WF of TCO substrate, enhance the hole extraction and transport, and improve the perovskite quality and stability, which can

Table 1. The performance of inverted PSCs with different HSSAMs.

HSSAM	Deposition method	Device structure	PCE [%]	References
MC-43	Dip coating	ITO/HSSAM/MAPbI ₃ /PC ₆₀ BM/Ag	17.3	[34]
MeO-2PACz	Spin coating	ITO/HSSAM/Cs ₅ (MA _{1.7} FA _{8.3}) ₉₅ Pb(I ₈₃ Br ₁₇) ₃ /C ₆₀ /BCP/Cu	20.4	[44]
MeO-2PACz	Spin coating	ITO/HSSAM(NiO _x)/Cs _{0.05} (FA _{0.83} MA _{0.17}) _{0.95} Pb(I _{0.82} Br _{0.18}) ₃ /C ₆₀ /BCP/Ag	19.9	[80]
MeO-2PACz	Spray coating	ITO/HSSAM/MAPbI ₃ /C ₆₀ /BCP/Ag	20.8	[99]
MeO-2PACz	Dip coating	ITO/HSSAM(NiO)/perovskite/C ₆₀ /BCP/Cu	20	[81]
MeO-2PACz	Spin coating	ITO/HSSAM/Cs _{0.05} (FA _{0.83} MA _{0.17}) _{0.95} Pb(I _{0.83} Br _{0.17}) ₃ /C ₆₀ /BCP/Ag	21.3	[95]
MeO-2PACz	Spin coating	ITO/HSSAM/FA _{0.6} MA _{0.4} PbI ₃ /C ₆₀ /BCP/Cu	23.1	[40]
MeO-2PACz	Spin coating	ITO/HSSAM/Cs _{0.05} MA _{0.15} FA _{0.80} PbI ₃ /C ₆₀ /BCP/Ag	22.89	[96]
2PACz	Spin coating	ITO/HSSAM/Cs ₅ (MA _{1.7} FA _{8.3}) ₉₅ Pb(I ₈₃ Br ₁₇) ₃ /C ₆₀ /BCP/Cu	20.9	[44]
2PACz	Spin coating	ITO/HSSAM/Cs _{0.05} MA _{0.15} FA _{0.80} PbI ₃ /C ₆₀ /BCP/Ag	22.93	[96]
2PACz	Vapor deposition	ITO/HSSAM/Cs _{0.05} MA _{0.22} FA _{0.73} Pb(I _{0.77} Br _{0.23}) ₃ /C ₆₀ /BCP/Ag	19.5	[101]
Me-4PACz	Spin coating	ITO/HSSAM/Cs _{0.05} (FA _{0.77} MA _{0.23}) _{0.95} Pb(I _{0.77} Br _{0.23}) ₃ /C ₆₀ /SnO ₂ /Ag	20.0	[45]
Me-4PACz	Spin coating	ITO/HSSAM/Cs _{0.05} (FA _{0.98} MA _{0.02}) _{0.95} Pb(I _{0.98} Br _{0.02}) ₃ /C ₆₀ /BCP/Ag	24.5	[97]
EADR04	Dip coating	ITO/HSSAM/Cs _{0.05} FA _{0.79} MA _{0.16} Pb(I _{0.84} Br _{0.16}) ₃ /C ₆₀ /BCP/Cu	20.6	[37]
TPT-P6	Spin coating	ITO/HSSAM/Cs _{0.05} MA _{0.12} FA _{0.83} Pb(I _{0.85} Br _{0.15}) ₃ /C ₆₀ /BCP/Ag	21.43	[67]
CbzNaph	Spin coating	ITO/HSSAM/Cs _{0.05} MA _{0.15} FA _{0.80} PbI ₃ /C ₆₀ /BCP/Ag	24.02	[96]
CbzNaph	Spin coating	ITO/HSSAM/Cs _{0.05} MA _{0.15} FA _{0.80} PbI ₃ /C ₆₀ /BCP/Ag	24.1	[47]
Br-2-EPSe	Spin coating	FTO/HSSAM/CsFAMA perovskite/C ₆₀ /BCP/Cu	22.73	[57]
Br-2EPT	Spin coating	FTO/HSSAM/Cs _{0.05} (FA _{0.92} MA _{0.08}) _{0.95} Pb(I _{0.92} Br _{0.08}) ₃ /C ₆₀ /BCP/Cu	22.44	[56]
MPA-Ph-CA	Spin coating	ITO/HSSAM/Cs _{0.05} (FA _{0.92} MA _{0.08}) _{0.95} Pb(I _{0.92} Br _{0.08}) ₃ /C ₆₀ /BCP/Ag	22.53	[74]
MPA-BT-CA	Dip coating	ITO/HSSAM/perovskite/C ₆₀ /ZrAcac/Ag	21.81	[68]
MTPA-BA	Spin coating	ITO/HSSAM/Cs _{0.05} (FA _{0.95} MA _{0.05}) _{0.95} Pb(I _{0.95} Br _{0.05}) ₃ /PCBM/C ₆₀ /BCP/Ag	22.62	[72]
3PATAT-C3	Spin coating	FTO/HSSAM/Cs _{0.05} FA _{0.80} MA _{0.15} PbI _{2.75} Br _{0.25} /C ₆₀ /BCP/Ag	23	[73]
PPA	Spin coating	ITO/HSSAM/Cs _{0.05} FA _{0.85} MA _{0.1} PbI ₃ PCBM/BCP/Ag	23.24	[66]
V1306 + C4	Dip coating	ITO/HSSAM/Cs _{0.05} (MA _{0.17} FA _{0.83}) _{0.95} Pb(I _{0.83} Br _{0.17}) ₃ /C ₆₀ /BCP/Cu	17.8	[29]
2PACz + MPA	Spin coating	FTO/HSSAM/Cs _{0.025} FA _{0.475} MA _{0.5} Sn _{0.5} Pb _{0.5} I _{2.925} Br _{0.075} /C ₆₀ /BCP/Ag	23.3	[84]
TPA-PT-C6 + CA-Br	Dip coating	ITO/HSSAM/MAPbI ₃ /PCBM/BCP/Ag	17.19	[90]
Br-2PACz + 4CzNH ₃ I	Dip coating	ITO/HSSAM/Cs _{0.25} FA _{0.75} Sn _{0.5} Pb _{0.5} I ₃ /C ₆₀ /BCP/Ag	19.45	[91]
Me-4PACz + 6dPA	Spin coating	ITO/HSSAM/FA _{0.75} Cs _{0.22} MA _{0.03} Pb(I _{0.82} Br _{0.15} Cl _{0.03}) ₃ /LiF/C ₆₀ /BCP/Ag	20.9	[88]
DC-PA + IAHA	Spin coating	ITO/HSSAM/Cs _{0.25} FA _{0.75} Sn _{0.5} Pb _{0.5} I ₃ /C ₆₀ /BCP/Ag	23.59	[85]
MPA-CPA	Spin coating	ITO/HSSAM/Cs _{0.05} (FA _{0.95} MA _{0.05}) _{0.95} Pb(I _{0.95} Br _{0.05}) ₃ /C ₆₀ /BCP/Ag	25.2	[30]

assist the fabrication of high-performance inverted PSCs and bring them closer to commercialization. Therefore, it is necessary to understand in depth the correlation between molecular structure and the microstructure and properties of HSSAM. However, a straightforward observation of HSSAM is a great challenge because of very low thickness of HSSAM. The most commonly used characterization methods in literatures, such as X-ray photoelectron spectroscopy, Fourier-transform infrared spectra, and the contact angel measurements, can only confirm the HSSAM formation on TCO substrate, but cannot reveal the microstructure and actual state of HSSAM. In this regard, advanced chemical imaging technology and interface-sensitive spectroscopy technique are required to fully characterize the microstructure and state of HSSAM, such as structural ordering, dynamic evolution, and defect.

HSSAMs used in inverted PSCs are located at the light-incident side of the device. Therefore, the light and thermal

stability of HSSAMs should be a great concern. HSSAMs are organic small molecules that can absorb some ultraviolet light. Therefore, depositing HSSAMs with excellent stability for inverted PSCs under operation condition is a challenge. Recently, it had been reported that the introduction of conjugated space group and enhancing the anchoring strength by screening the anchor groups could effectively enhance the stability of HSSAM in inverted PSCs.^[67,74] These results revealed that the molecule design could be a promising and feasible strategy for promoting the stability of HSSAMs in inverted PSCs.

Although the development of HSSAMs in inverted PSCs faces several challenges and is in its infancy, the great achievement in the past five years and the versatility of HSSAMs enable them to show a great potential in enhancing the performance of inverted PSCs and promoting the development of tandem solar cells. It is certain that the HSSAM strategy must make a remarkable

contribution to the development of both single junction and tandem PSCs in the future.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

hole-selective layer, inverted perovskite solar cells, self-assembled monolayers

Received: December 1, 2023

Revised: January 14, 2024

Published online: January 29, 2024

- [1] J. Correa-Baena, M. Saliba, T. Buonassisi, M. Grätzel, A. Abate, W. Tress, A. Hagfeldt, *Science* **2017**, 358, 739.
- [2] J. Kim, J. Lee, H. Jung, H. Shin, N. Park, *Chem. Rev.* **2020**, 120, 7867.
- [3] C. Ashworth, *Nat. Rev. Mater.* **2021**, 6, 293.
- [4] M. Kim, J. Jeong, H. Lu, T. Lee, F. Eickemeyer, Y. Liu, I. W. Choi, S. Choi, Y. Jo, H. Lee, N. An, S. Cho, W. Tress, S. M. Zakeeruddin, A. Hagfeldt, J. Kim, M. Grätzel, D. Kim, *Science* **2022**, 375, 302.
- [5] J. Park, J. Kim, H. Yun, M. Paik, E. Noh, H. Mun, M. Kim, T. Shin, S. Seok, *Nature* **2023**, 616, 724.
- [6] Best Research-Cell Efficiency Chart (NREL), <https://www.nrel.gov/pv/cell-efficiency.html> (accessed: October 2023).
- [7] J. Yoo, G. Seo, M. Chua, T. Park, Y. Lu, F. Rotermund, C. Moon, N. Jeon, J. Correa-Baena, M. Bawendi, J. Seo, *Nature* **2021**, 590, 587.
- [8] H. Min, D. Lee, J. Kim, M. Paik, Y. Kim, G. Kim, K. Kim, K. Lee, J. Kim, M. Kim, T. Shin, S. Seok, *Nature* **2021**, 598, 444.
- [9] Y. Zhao, F. Ma, Z. Qu, S. Yu, T. Shen, S. Deng, X. Peng, Y. Yuan, X. Zhang, J. You, *Science* **2022**, 377, 531.
- [10] M. Stollerfoht, C. Wolff, J. Márquez, S. Zhang, C. Hages, D. Rothhardt, S. Albrecht, P. L. Burn, P. Meredith, T. Unold, D. Neher, *Nat. Energy* **2018**, 3, 847.
- [11] M. Stollerfoht, P. Caprioglio, C. M. Wolff, J. A. Márquez, J. Nordmann, S. Zhang, D. Rothhardt, L. Kegelmann, F. Zu, S. Albrecht, M. Saliba, T. Unold, D. Neher, *Energy Environ. Sci.* **2019**, 12, 2778.
- [12] E. Aktas, N. Phung, H. Köbler, D. González, M. Méndez, R. Wenisch, I. Lauermann, A. Abate, E. Palomares, *Energy Environ. Sci.* **2021**, 14, 3976.
- [13] W. Kong, W. Li, C. Liu, H. Liu, J. Miao, W. Wang, S. Chen, M. Hu, D. Li, A. Amini, *ACS Nano* **2019**, 13, 4049.
- [14] Y. Deng, S. Xu, S. Chen, X. Xiao, J. Zhao, J. Huang, *Nat. Energy* **2021**, 6, 633.
- [15] S. Wu, J. Zhang, Z. Li, D. Liu, M. Qin, S. Cheung, X. Lu, D. Lei, S. So, Z. Zhu, A. Jen, *Joule* **2020**, 4, 1248.
- [16] S. Wu, Z. Li, M. Q. Li, Y. Diao, F. Lin, T. Liu, J. Zhang, P. Tieu, W. Gao, F. Qi, X. Pan, Z. Xu, Z. Zhu, A. Jen, *Nat. Nanotechnol.* **2020**, 15, 934.
- [17] J. Xu, C. Boyd, Z. Yu, A. Palmstrom, D. Witter, B. Larson, R. France, J. Werner, S. Harvey, E. Wolf, W. Weigand, S. Manzoor, J. Berry, J. Luther, Z. Holman, M. McGehee, *Science* **2020**, 367, 1097.
- [18] Z. Li, B. Li, X. Wu, S. A. Sheppard, S. Zhang, D. Gao, N. Long, Z. Zhu, *Science* **2022**, 376, 416.
- [19] Q. Tan, Z. Li, G. Lou, X. Zhang, G. Chen, G. He, J. Wang, J. Xiu, H. Yi, T. Chen, Z. He, *Nature* **2023**, 620, 545.
- [20] C. Li, X. Wang, E. Bi, F. Jiang, S. Park, L. Chen, Y. Liu, C. Grice, J. Chung, T. Zhu, B. Chen, F. Fu, D. Ginger, Z. Song, E. Sargent, Y. Yan, *Science* **2023**, 379, 690.
- [21] Y. Yao, C. Cheng, C. Zhang, H. Hu, K. Wang, S. De Wolf, *Adv. Mater.* **2022**, 34, 2203794.
- [22] W. Nie, H. Tsai, J. Blancon, H. F. Liu, C. Stoumpos, M. Kepenekian, O. Durand, C. Katan, S. Tretiak, M. Kanatzidis, J. Even, A. Mohite, *Adv. Mater.* **2018**, 30, 1703879.
- [23] Y. Jiang, J. Wang, H. Zai, N. Li, H. Lu, F. Wang, Z. Bi, H. Xie, Q. Wang, W. Ma, Y. Tu, H. Zhou, X. Zhan, *J. Am. Chem. Soc.* **2022**, 144, 5400.
- [24] T. Liu, K. Chen, R. Zhu, Q. Gong, *Adv. Energy Mater.* **2016**, 6, 1600457.
- [25] X. Yin, Y. Guo, H. Xie, W. Que, L. Kong, *Sol. RRL* **2019**, 3, 1900001.
- [26] H. Rao, S. Ye, W. Sun, W. Yan, Y. Li, H. Peng, Z. Liu, Z. Bian, Y. Li, C. Huang, *Nano Energy* **2016**, 27, 51.
- [27] H. Cheng, Y. Li, Y. Zhong, *Mater. Chem. Front.* **2023**, 7, 3958.
- [28] S. Wang, H. Guo, Y. Wu, *Mater. Futures* **2023**, 2, 012105.
- [29] A. Magomedov, A. Al-Ashouri, S. Strazdaite, G. Niaura, T. Malinauskas, S. Albrecht, V. Getautis, *Adv. Energy Mater.* **2018**, 8, 1801892.
- [30] S. Zhang, F. Ye, X. Wang, R. Chen, H. Zhang, L. Zhan, X. Ji, S. Liu, Z. Ning, D. Neher, L. Han, H. Tian, W. Chen, M. Stollerfoht, L. Zhang, W. Zhu, Y. Wu, *Science* **2023**, 380, 404.
- [31] R. Nuzzo, D. Allara, *J. Am. Chem. Soc.* **1983**, 105, 4481.
- [32] J. Love, L. Estroff, J. Kriebel, R. Nuzzo, G. Whitesides, *Chem. Rev.* **2005**, 105, 1103.
- [33] S. Kim, S. Cho, S. Byeon, X. He, H. Yoon, *Adv. Energy Mater.* **2020**, 10, 2002606.
- [34] E. Yalcin, M. Can, C. Rodriguez-Seco, E. Aktas, R. Pudi, W. Cambarau, S. Demic, E. Palomares, *Energy Environ. Sci.* **2019**, 12, 230.
- [35] X. Yin, L. Guan, J. Yu, D. Zhao, C. Wang, Q. An, B. Zhou, Y. Yu, C. Grice, R. Awni, R. Ellingson, Y. Yan, W. Tang, *Nano Energy* **2017**, 40, 163.
- [36] Q. Wang, E. Mosconi, C. Wolff, J. Li, F. De Angelis, R. Grisorio, A. Abate, *Adv. Energy Mater.* **2019**, 9, 1900990.
- [37] E. Aktas, N. Phung, H. Köbler, D. González, M. Méndez, I. Kafedjiska, S. Turren-Cruz, A. Abate, E. Palomares, *Energy Environ. Sci.* **2021**, 14, 3976.
- [38] C. Huang, W. Fu, C. Li, Z. Zhang, W. Qiu, M. Shi, A. Jen, H. Chen, *J. Am. Chem. Soc.* **2016**, 138, 2528.
- [39] M. Saliba, S. Orlandi, T. Matsui, S. Aghazada, P. Cao, R. Scopelliti, E. Mosconi, K. Dahmen, F. De Angelis, A. Hagfeldt, M. Graetzel, M. K. Nazeeruddin, *Nat. Energy* **2016**, 1, 15017.
- [40] K. Almasabi, X. Zheng, B. Turedi, A. Alsalloum, K. Kotsovos, A. Jamal, I. Gereige, O. Mohammed, O. Bakr, *ACS Energy Lett.* **2023**, 8, 950.
- [41] C. Huang, W. Fu, C. Z. Li, Z. Zhang, W. Qiu, M. Shi, P. Heremans, A. Jen, H. Chen, *J. Am. Chem. Soc.* **2016**, 138, 2528.
- [42] S. Park, S. Jeon, I. Lee, J. Zhang, H. Jeong, J. Park, J. Bang, T. Ahn, H. Shin, B. Kim, H. Park, *J. Mater. Chem. A* **2017**, 5, 13220.
- [43] I. Lange, S. Reiter, M. Pätz, A. Zykov, A. Nefedov, J. Hildebrandt, S. Hecht, S. Kowarik, G. Heimel, D. Neher, *Adv. Funct. Mater.* **2014**, 24, 7014.
- [44] A. Al-Ashouri, A. Magomedov, M. Ros, M. Jošt, M. Talaikis, G. Chistiakov, R. Schlattmann, B. Rech, T. Malinauskas, G. Niaura, V. Getautis, S. Albrecht, *Energy Environ. Sci.* **2019**, 12, 3356.
- [45] I. Levine, A. Al-Ashouri, A. Musiienko, H. Hempel, A. Magomedov, A. Drevilkauskaitė, V. Getautis, D. Menzel, K. Hinrichs, T. Unold, S. Albrecht, T. Dittrich, *Joule* **2021**, 5, 2919.
- [46] A. Al-Ashouri, E. Köhnen, B. Li, A. Magomedov, H. Hempel, P. Caprioglio, D. Menzel, M. Grischek, M. Jošt, L. Korte, A. Abate, B. Stannowski, T. Unold, V. Getautis, S. Albrecht, *Science* **2020**, 370, 1300.
- [47] W. Jiang, F. Li, M. Li, F. Qi, F. Lin, A. Jen, *Angew. Chem., Int. Ed.* **2022**, 61, e202213560.

- [48] F. Ali, C. Roldán-Carmona, M. Sohail, M. K. Nazeeruddin, *Adv. Energy Mater.* **2020**, *10*, 2002989.
- [49] G. Wang, D. Wang, J. Chang, W. Xiang, F. Meng, *Sol. RRL* **2023**, *7*, 202300481.
- [50] a) W. Wu, Y. Liu, D. Zhu, *Chem. Soc. Rev.* **2010**, *39*, 1489; b) W. Jiang, Z. Liu, D. Zhu, W. Zheng, L. Chen, X. Zhang, G. Zhang, Y. Yi, L. Jiang, D. Zhang, *Angew. Chem., Int. Ed.* **2021**, *60*, 10700.
- [51] S. Casalini, C. Bortolotti, F. Leonardi, F. Biscarini, *Chem. Soc. Rev.* **2017**, *46*, 40.
- [52] D. Shinde, J. Salunke, N. Candeias, F. Tinti, M. Gazzano, P. Wadgaonkar, A. Priimagi, N. Camaioni, P. Vivo, *Sci. Rep.* **2017**, *7*, 46268.
- [53] Z. Dai, S. Yadavalli, M. Chen, A. Abbaspourtamijani, Y. Qi, N. Padture, *Science* **2021**, *372*, 618.
- [54] X. Liu, Y. Cheng, B. Tang, Z. Yu, M. Li, F. Lin, S. Zhang, Y. Zhang, J. Ouyang, H. Gong, *Nano Energy* **2020**, *71*, 104556.
- [55] X. Li, W. Zhang, X. Guo, C. Lu, J. Wei, J. Fang, *Science* **2022**, *375*, 434.
- [56] A. Ullah, K. Park, H. Nguyen, Y. Siddique, S. Shah, H. Tran, S. Park, S. Lee, S. Ahn, I. Jeong, Y. Park, S. Hong, *Adv. Energy Mater.* **2022**, *12*, 2103175.
- [57] A. Ullah, K. Park, Y. Lee, S. Park, A. Faheem, H. Nguyen, Y. Siddique, K. Lee, I. Jeong, S. Cho, B. Kim, Y. Park, S. Hong, *Adv. Funct. Mater.* **2022**, *32*, 2208793.
- [58] G. You, L. Li, S. Wang, J. Cao, L. Yao, W. Cai, Z. Zhou, K. Li, Z. Lin, H. Zhen, Q. Ling, *Adv. Energy Mater.* **2022**, *12*, 2102697.
- [59] J. Wang, K. Liu, L. Ma, X. Zhan, *Chem. Rev.* **2016**, *116*, 14675.
- [60] E. Aktas, R. Pudi, N. Phung, R. Wenisch, L. Gregori, D. Meggiolaro, F. De Angelis, A. Abate, E. Palomares, *ACS Appl. Mater. Interfaces* **2022**, *14*, 17461.
- [61] Z. Yao, M. Zhang, R. Li, L. Yang, Y. Qiao, P. Wang, *Angew. Chem., Int. Ed.* **2015**, *54*, 5994.
- [62] L. Zhang, X. Yang, W. Wang, G. Gurzadyan, J. Li, X. Li, J. An, Z. Yu, H. Wang, A. Hagfeldt, L. Sun, *ACS Energy Lett.* **2019**, *4*, 943.
- [63] Y. Wang, W. Chen, L. Wang, B. Tu, T. Chen, B. Liu, K. Yang, C. Koh, X. Zhang, H. Sun, G. Chen, H. Woo, Z. He, X. Guo, *Adv. Mater.* **2019**, *31*, 1902781.
- [64] Y. Wang, Q. Liao, J. Chen, W. Huang, X. Zhuang, Y. Tang, X. Zhang, M. Su, Z. He, T. Marks, A. Facchetti, X. Guo, *J. Am. Chem. Soc.* **2020**, *142*, 16632.
- [65] A. Ok, B. Jo, S. Somasundaram, H. Woo, D. Lee, Z. Li, B. Kim, J. Kim, Y. Song, T. Ahn, S. Park, H. Park, *Nat. Commun.* **2018**, *9*, 4537.
- [66] R. Guo, X. Zhang, X. Zheng, L. Li, M. Li, Y. Zhao, S. Zhang, L. Luo, Y. Cui, Y. Rong, H. Zeng, *Adv. Funct. Mater.* **2023**, *33*, 2211955.
- [67] E. Li, C. Liu, H. Lin, X. Xu, S. Liu, S. Zhang, M. Yu, X. Cao, Y. Wu, W. Zhu, *Adv. Funct. Mater.* **2021**, *31*, 2103847.
- [68] Q. Liao, Y. Wang, Z. Zhang, K. Yang, Y. Shi, K. Feng, J. Huang, P. Gao, X. Guo, *J. Energy Chem.* **2022**, *68*, 875.
- [69] C. Chen, X. Wang, Z. Li, X. Du, Z. Shao, X. Sun, D. Liu, C. Gao, B. Zhang, G. Cui, S. Pang, *Angew. Chem., Int. Ed.* **2022**, *61*, e202113932.
- [70] M. P. de Jong, M. de Voigt, *Appl. Phys. Lett.* **2000**, *77*, 2255.
- [71] T. Bejtual, K. Ramji, A. Kessman, K. Sierros, D. Cairns, *Mater. Chem. Phys.* **2012**, *132*, 395.
- [72] H. Guo, C. Liu, H. Hu, S. Zhang, X. Ji, Z. Ning, W. Zhu, H. Tian, Y. Wu, *Natl. Sci. Rev.* **2023**, *10*, nwad057.
- [73] M. Truong, T. Funasaki, L. Ueberricke, W. Nojo, N. Sekiguchi, S. Hira, L. Xie, T. Nakamura, S. Iikubo, H. Yoshida, Y. Shimakawa, T. Suzuki, A. Wakamiya, *J. Am. Chem. Soc.* **2023**, *145*, 7528.
- [74] S. Zhang, R. Wu, C. Mu, Y. Wang, L. Han, Y. Wu, W. Zhu, *ACS Mater. Lett.* **2022**, *4*, 1976.
- [75] M. Cygan, T. Dunbar, J. Arnold, L. Bumm, N. Shedlock, T. Burgin, L. Jones, D. Allara, J. Tour, P. Weiss, *J. Am. Chem. Soc.* **1998**, *120*, 2721.
- [76] M. Losego, J. Guske, A. Efremenko, J. Maria, S. Franzen, *Langmuir* **2011**, *27*, 11883.
- [77] A. Ulman, *Chem. Rev.* **1996**, *96*, 1533.
- [78] Y. Lin, Y. Firdaus, F. Isikgor, M. Nugraha, E. Yengel, G. Harrison, R. Hallani, A. El-Labban, H. Faber, O. Bakr, I. McCulloch, S. Wolf, L. Tsetseris, T. Anthopoulos, *ACS Energy Lett.* **2020**, *5*, 2935.
- [79] H. Ma, O. Acton, G. Ting, J. Ka, H. Yip, N. Tucker, R. Schofield, A. Jen, *Appl. Phys. Lett.* **2008**, *92*, 113303.
- [80] J. Sun, C. Shou, J. Sun, X. Wang, J. Wu, X. Yang, J. Sheng, B. Yan, J. Ye, *Sol. RRL* **2021**, *5*, 2100663.
- [81] N. Phung, M. Verheijen, A. Todinova, A. Al-Ashouri, X. Li, A. Abate, S. Albrecht, M. Creatore, *ACS Appl. Mater. Interfaces* **2022**, *14*, 2166.
- [82] A. Khassanov, H. Steinruck, T. Schmaltz, A. Magerl, M. Halik, *Acc. Chem. Res.* **2015**, *48*, 1901.
- [83] A. Rumpel, M. Novak, J. Walter, B. Braunschweig, M. Halik, W. Peukert, *Langmuir* **2011**, *27*, 15016.
- [84] G. Kapil, T. Bessho, Y. Sanehira, S. Sahamir, M. Chen, D. Nomura, K. Nishimura, M. Kamarudin, Q. Shen, H. Segawa, S. Hayase, *ACS Energy Lett.* **2022**, *7*, 966.
- [85] X. Deng, F. Qi, F. Li, S. Wu, F. Lin, Z. Zhang, Z. Guan, Z. Yang, C. Lee, A. Jen, *Angew. Chem., Int. Ed.* **2022**, *61*, e202203088.
- [86] J. Chen, X. Zhao, S.-G. Kim, N.-G. Park, *Adv. Mater.* **2019**, *31*, 1902902.
- [87] L. Zuo, Q. Chen, N. De Marco, Y. Hsieh, H. Chen, P. Sun, S. Chang, H. Zhao, S. Dong, Y. Yang, *Nano Lett.* **2017**, *17*, 269.
- [88] A. Al-Ashouri, M. Marčinkas, E. Kasparavičius, T. Malinauskas, S. Albrecht, M. McGehee, A. Magomedov, *ACS Energy Lett.* **2023**, *8*, 898.
- [89] K. Hossain, A. Kulkarni, U. Bothra, B. Klingebiel, T. Kirchartz, M. Saliba, D. Kabra, *ACS Energy Lett.* **2023**, *8*, 3860.
- [90] E. Li, E. Bi, Y. Wu, W. Zhang, L. Li, H. Chen, L. Han, H. Tian, W. Zhu, *Adv. Funct. Mater.* **2019**, *29*, 1909509.
- [91] M. Pitaro, J. E. S. Alonso, L. Mario, D. Romero, T. Zaharia, E. M. J. Johansson, R. Chiechi, M. Loi, *Adv. Funct. Mater.* **2023**, *33*, 2306571.
- [92] J. Bardecker, H. Ma, T. Kim, F. Huang, Y. Cheng, G. Ting, A. Jen, *Adv. Funct. Mater.* **2008**, *18*, 3964.
- [93] P. Hotchkiss, S. Jones, S. Paniagua, A. Sharma, B. Kippelen, N. Armstrong, S. Marder, *Acc. Chem. Res.* **2012**, *45*, 337.
- [94] Y. Lin, Y. Firdaus, F. Isikgor, M. Nugraha, R. Hallani, A. El-Labban, H. Faber, C. Ma, O. Bakr, S. De Wolf, L. Tsetseris, T. Anthopoulos, *ACS Energy Lett.* **2020**, *5*, 2935.
- [95] M. Wu, X. Li, Z. Ying, Y. Chen, X. Wang, X. Guo, J. Sun, C. Shou, X. Yang, J. Ye, *Adv. Funct. Mater.* **2023**, *33*, 2304708.
- [96] M. Liu, L. Bi, W. Jiang, Z. Zeng, S. Tsang, F. R. Lin, A. Jen, *Adv. Mater.* **2023**, *35*, 2304415.
- [97] X. Zheng, Z. Li, Y. Zhang, M. Chen, T. Liu, C. Xiao, D. Kuciauskas, A. Magomedov, S. Harvey, Z. Dai, A. Kirmani, K. Graham, Y. Yan, M. Nazeeruddin, M. McGehee, Z. Zhu, J. Luther, *Nat. Energy* **2023**, *8*, 462.
- [98] M. Eslamian, *Coatings* **2014**, *4*, 60.
- [99] E. Cassella, E. Spooner, T. Thornber, M. O'Kane, T. Catley, O. Game, D. Lidzey, *Adv. Sci.* **2022**, *9*, 2104848.
- [100] L. Huo, T. Liu, X. Sun, Y. Cai, A. J. Heeger, Y. Sun, *Adv. Mater.* **2015**, *27*, 2938.
- [101] A. Farag, T. Feeney, I. Hossain, F. Schackmar, M. Ruiz-Preciado, M. Hentschel, D. B. Ritzer, A. Diercks, Y. Li, B. Laufer, R. Singh, U. Starke, U. Paetzold, *Adv. Energy Mater.* **2023**, *13*, 2203982.



Guiqiang Wang obtained his Ph.D. degree from the Institute of Chemistry, Chinese Academy of Science in 2004. Then, he worked as an associate professor at Shandong University of Technology. He joined the Department of New Energy of Bohai University as a professor in 2014. His research interests mainly focus on energy conversion and storage devices, including perovskite solar cells, dye-sensitized solar cells, and supercapacitors.



Fanning Meng received her Ph.D. degree from the Dalian University of Technology. Now she is working at Bohai University. Her interests include researching on preparation and application of organic chemical materials, polymeric materials, and perovskite solar cells.



Wanchun Xiang received his M.S. from the Shaanxi Normal University in 2005 and his Ph.D. from the Institute of Chemistry, Chinese Academy of Sciences in 2010. After three years of postdoctoral research with Prof. Leone Spiccia at the Department of Chemistry, Monash University (Australia), he joined the State Key Laboratory of Silicate Materials for Architectures, Wuhan University of Technology as an associate professor in 2014. He has been a professor at the Shaanxi Normal University since 2020. His research interests include optoelectronic materials and their applications in photovoltaic devices, such as dye-sensitized solar cells and perovskite solar cells.